A STATISTICAL ANALYSIS OF WILDLAND AND PRESCRIBED FIRES IN THE CHANGING CALIFORNIA ENVIRONMENT

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Master of Science
In
Statistical Data Science

by

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CERTIFICATION OF APPROVAL

I certify that I have read *A STATISTICAL ANALYSIS OF WILDLAND AND PRESCRIBED FIRES IN THE CHANGING CALIFORNIA ENVIRONMENT* by Sequoia Andrade and that in my opinion this work meets the criteria for approving a thesis submitted in partial fulfillment of the requirements for the degree: Master of Science in Statistical Data Science at San Francisco State University.

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With climate change-driven large, destructive wildfires becoming more frequent, especially in California, using data to identify effective or ineffective prevention strategies, such as prescribed burns, can inform future prevention strategies. In this work, we present a statistical analysis on a new dataset created by fusing wildfire, prescribed fire, and climate data. Descriptive and inferential statistics, time series analysis, and regression analysis are applied to the dataset to analyze if prescribed burns are associated with reduced wildfire size while accounting for climate variables. When grouped across the entire state with a monthly time scale, results indicate prescribed burns do not impact the number of wildfire acres burned. However, the impact of prescribed burns is significant when considering individual fires, even when controlling for climate variables. Climate variables have a stronger impact on wildfire size, though prescribed burns are associated with slightly smaller wildfires.

I certify that the Abstract is a correct representation of the content of this thesis.

Chair, Thesis Committee Date

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Chapter 1

Introduction

Wildfires are increasing in size, frequency, and severity in California, with 60% of the ten largest fires in state history taking place in from 2020-2023 [11] and 50% of the ten most destructive fires taking place from 2018-2023 [12]. In 2020, California experienced a record-breaking wildfire season with over 8,000 fires burning over 4.3 million acres in total, resulting in 11,000 structures destroyed, 33 fatalities, and approximately 12 billion dollars in damage and suppression costs [41, 9]. Existing research indicates decreased humidity, in part attributed to human activities [1], alongside an increasingly populated wildland-urban interface [32] has enabled the destructive evolution of fires seen present day.

With climate change-driven large, destructive wildfires becoming more frequent, especially in California, using data to identify effective or ineffective prevention strategies, such as prescribed burns, can inform future prevention strategies. Prescribed burns are intentional, controlled fires performed by forest management or-

ganizations to burn existing vegetation in hopes of mitigating the spread of any potential wildfires by reducing fuel. This practice was commonly performed by indigenous people for generations to cultivate land and provide protection. There exists a large body of research demonstrating the effectiveness of prescribed burns for mitigating the effects of large wildfires; however, much of this research was performed in the previous century and did not account for climate change's impact on prescribed burn efficacy. Numerous datasets have been made public in recent years, including detailed tree-level data from national parks in addition to incident reports from large wildfire events. Statistical data science methods provide tools for making sense of the immense heterogeneous data relevant to wildfires.

Throughout this report statistical methods are applied to wildfire-relevant data to better understand the complex interplay between wildfires, prescribed fires, and climate factors in California from 2016-2023. This problem is inherently complex due to data variation, lack of paired data, and spatio-temporal components. An overview of relevant research is provided, as well as a discussion of available datasets. A custom dataset suitable for the analysis is created through aggregating disparate sources. Exploratory analysis including descriptive and inferential statistics is provided. Multiple analyses are performed to examine the relationship between wildfires, prescribed fires, and climate factors through different lenses. First, a time series analysis is performed to identify relevant correlations and time lags. Next, a spatio-temporal analysis is performed. Regression analysis is applied to the data

to identify which variables are most related to wildfire size, followed by a principle component analysis to summarize the variation of the data. A discussion is provided to conclude the report.

Chapter 2

Literature Review

2.1 Prescribed Burns

The efficacy of prescribed burns has been studied many times of the years, with a recent meta-analysis identifying metrics such as burned area and wildfire frequency as useful for determining the impact of prescribed burns [21]. While prescribed burns are a known effective method for fuel management to reduce the severity of wildfires, barriers such as risk and resources prevent their widespread use [28]. Prescribed burns can only performed only ideal conditions, with recent research finding the number of viable burn days per year dropping over time [4]. Social views of prescribed fire tend to be negative, with citizens citing potential environmental pollution and contamination as side effects. However, research indicates the effect of prescribed burns on the environment is temporary [5]. Additional research has found some variables, such as elevation, mediate the impact prescribed burns have on water

contamination [7]. Prescribed burns have been found to not affect near by fungus communities [40]. Both prescribed burn landscape management and community preparation are necessary to minimize the amount of acres burned [3].

2.2 Climate Change and Wildfires

The relationship between climate factors and wildfires has been studied thoroughly, leading to a wide body of research using different methods. Recently, cross-correlation, as well as categorical methods (i.e., chi-squared), have been used to study the link between climate factors and wildfire events in Africa [22]. Additional research has used regional cross-correlation and auto-correlation to identify trends in simultaneously occurring fires, with low fuel moisture and lightning contributing to high numbers of simultaneous fires from 1984-2015 [30]. A cross-correlational study of humidity, precipitation, wind, and temperature has been used to identify a strong relationship between humidity, precipitation, and wildfire [24]. Empirical orthogonal function analysis has been used to investigate the relationship between atmospheric variability and wildfires, with findings indicating that drought impacts wildfire occurrence but is moderated by forest composition[19]. Further research using power-law statistics identified different thresholds for tail events, with drought impacting large fires and wind impacting smaller fires [36]. The relationship between regional weather patterns impacting wildfire occurrence have been studied using cross-correlation, with some regions experiencing heightened fire due to a lack

of rainfall and others experiencing greater burned area during heat waves after a rainy season [34, 33]. Similarly, research on Brazilian wildfires has found some climatic variables do not impact wildfires across subzones [17]. Targeted research of wildfires in Idaho identified differences in the Evaporative Demand Drought Index and Normalized Difference Vegetation Index between intense fire years and low fire years by examining moving averages and statistical distributions [18]. A Maxenet (maximum entropy) has also been used for fire prediction on various regions and spatial resolutions [6].

Chapter 3

Data Description

3.1 Dataset Candidates

Prior to constructing a dataset suitable for this study, existing datasets are explored and evaluated. There are five classes of datasets found to be relevant to this research: prescribed burns in CA, CAL-Fire reported fires, the ICS-209-PLUS v2, national park-level datasets from the USGS, and annual region-based weather data from NOAA. Table 3.1 provides an overview of each dataset, with descriptions, years of data, and relevant variables. Corresponding methods used to analyze some of these datasets are available in Table 3.2. Each dataset can be analyzed separately, while integrated datasets can be analyzed together. For example, the relationship between prescribed burns and large wildfires may be mediated by the weather of the given year, hence NOAA climate data is needed to fully analyze prescribed burn efficacy.

3.2 Prescribed Fires, Wildfires, and Climate Dataset

The dataset used in this project was manually created by bridging three main data sources: the Calfire record of wildfire incidents [10], the record of prescribed burn incidents [8], and climate data collected by NOAA [29]. CalFire provides a detailed record of all wildfire incidents in the state of California [1], with a row for each fire incident and data for acres burned, location, start date, and end date [10]. Initial exploration found that the end date is unreliable with many reports from 2017 grouped into ending in 2018, thus the start date is used instead for this analysis. The prescribed burn dataset also comes from CalFire and spans from 1950 to the present [2] [8]. Each row is a prescribed burn incident and includes the date, location, geographic polygon, and acres burned. Climate data is gathered primarily from NOAA [3] [29]. Monthly data is gathered for the state of California across the available variables: soil moisture, temperature average, temperature max, temperature min, cooling days, heating days, precipitation, and drought indexes. Each variable contains monthly average and anomaly data.

The resulting dataset used in this project contains a row for wildfire incident, spanning from 2013 to 2022, with variables bridged from the NOAA and prescribed fire data. Columns include climate data (precipitation, soil moisture, temperature, and drought index) for the month before the fire and temporal-spatial prescribed

[1] available at `https://incidents.fire.ca.gov/imapdata/mapdataall.csv`

[2] available at `https://data.ca.gov/dataset/prescribed-burns`

[3] available at `https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/statewide/time-series`

fire information. An overview of the variables is provided in Table 3.3. For each incident, we identify if a prescribed fire occurred within a specified time and spatial radius and record the sum of prescribed acres burned. This results in columns for each combination for 3-month, 6-month, 12-month, 24-month, and 36-month time frames for spatial radii of 5 km, 10 km, 25 km, 50 km, and 100 km. A new variable categorizing fires into one of eight regions (Central Valley, Central Coast, Shasta Cascades, Sierras, Dessert, South Coast, North Coast, and Multiple) is created from the county identified for each incident. Fire events with invalid latitude and longitude are removed. The resulting data set has 2041 rows and 70 columns. Additionally, a monthly version of the data is used for time series analysis, which includes the monthly sum of wildfire acres, prescribed fire acres, and monthly average climate variables across the state.

Table 3.1: Dataset descriptions, variables, and date information.

Title	Description	Dates	Variables
ICS-209-PLUS[37]	Dataset of incident and situation reports of large wildfire events	2006-2020	narrative, acres burned, structures damaged, cost, etc
CALFire record	dataset of all wildfires managed by calfire	2013-2022	acres, location, unit, date, county
SNFDP[13]	Intended for studying tree mortality, fire, and bark beetle interactions in Sequoia National Park. Data points are one per tree.	1990-2019	fire year, species, mortality year, crown volume scorch, neighboring trees, basal area
Treemortality[16]	Tree mortality data from extreme drought (2014-2017) and comparison years	2004-2007, 2014-2017	Plots, mortality year, diseases, insects
Prescribed Burns[8]	Dataset of geographic prescribed burns in California	1950-present	year, agency, acres, unit, shape
ECOSTRESS	Satellite-based data source hosted by JPL that measures plant stress	2018 - present	surface temperature and emissivity
Seed data [14]	long term seed and tree data from sequoia and yosemite	2000-2018	individual tree data overtime, seed data, plot data
Seed data early [15]	long term seed and tree data from sequoia and yosemite	1999-2008	individual tree data overtime
forest structure [26]	individual tree and plot information	unsure, seems to be a one time count	unsure
Precipitation data [2]	Monthly precipitation for the globe in 2.5 degree resolution grid in mm/day	1979-present	precipitation
Temperature anomaly [39]	Monthly temperature deviation from the long-term mean	1880-present	temperature
Soil Moisture [31]	Provides monthly soil moisture measurements	1981-2020	monthly mean, long term mean

Table 3.2: Methods and models information.

Model	Variables	Dataset	Source
Generalized Linear Mixed Models (GLMM)	Binomial family and logit, fire area as random effect	SNFDP	[13]
		N/A	[35]
Generalized Linear Model with Binomial distribution	Mortality 3-years post fire	N/A	[35]
Proportion tests (Fischer's)	Mortality in various species, beetles, drought	Tree Mortality dataset	[38]
Chi-squared	Burn vs Unburned, Mortality between scar groupings	N/A	[23]
Cross-correlation	climate variables vs wildfire variables	numerous	
Markov Chain	Unburned, burn, mortality	unknown	

Table 3.3: Overview of variables in the dataset.

Variable	Source	Description	Range	Units	Symbol
Acres Burned	CALfire[10]	Wildfire acres burned	0+	Acres	log_acres
Treatment Acres	CALfire[8]	Prescribed fire acres burned	0+	Acres	rx_acres
Precipitation	NOAA[29]	Monthly precipitation averaged across the county of the wildfire	0+	inches	pcp
Temperature	NOAA[29]	Average, minimum, and maximum temperature averaged across the county of the wildfire	0+	F	temp
Palmer Index[29]	NOAA	Drought Severity index, hydrological drought index, modified drought index, Z-index	[-10, 10]	NA	zndx
Maximum Wind Speed	Open Meteo [20, 42]	wind speed	0+	km/h	max_wind

Chapter 4

Data Exploration

4.1 Descriptive Statistics

Prior to analysis, the dataset is analyzed using descriptive statistics to understand underlying trends. Tables 4.1 and 4.2 show summaries for continuous variables in the wildfire and prescribed fire data respectively. From Table 4.1 we can see there are 2041 wildfire events with an average size of about 5,038 acres. The largest fire in the dataset is over one million acres, while the smallest is zero acres. From the quartiles, 75% of fires are 424 acres or less. Wildfires occur throughout the years and months, with at least 50% of events occurring between June and August. For prescribed fires, there are 4,123 total events with an average size of 95.359 acres. The largest prescribed fire was 7,006 acres, while the smallest is zero acres. Here 75% of fires are less than 76 acres in size. Half of prescribed burns occur between January and June, with another 25% occuring between November and December. Most

prescribed burns in the data set occur after 2018. Table 4.3 shows the summaries of categorical data from wild and prescribed fires. Wildfires occur most often in Riverside county and the Central Valley region, while prescribed fires occure the most in Siskiyou county and the Shasta Cascades region. Since California has fifty-eight counties, prescribed fires are not performed in every county (51), while wildfires have spanned multiple counties (97 unique counties and combinations).

Table 4.1: Descriptive statistics of numeric calfire data.

metric	acres burned	longitude	latitude	year	month
count	2041	2041	2041	2041	2041
mean	5038.136	-120.281	37.293	2018	7.233
std	39490.062	2.022	2.582	2.417	1.821
min	0	-124.200	32.558	2013	1
25%	35	-121.854	34.919	2017	6
50%	100	-120.705	37.494	2018	7
75%	424	-118.739	39.355	2020	8
max	1032648	-114.484	42.048	2022	12

Table 4.2: Descriptive statistics of prescribed burn numerical data.

metric	acres burned	longitude	latitude	year	month
count	4123	4123	4123	4123	4123
mean	95.359	38.854	-120.792	2019.028	7.041
std	343.280	2.360	1.775	2.152	3.870
min	0	32.583	-124.359	2013	1
25%	4.076	37.340	-121.992	2018	4
50%	22.214	39.384	-120.863	2019	6
75%	76	40.774	-119.879	2020	11
max	7006	41.999	-115.827	2022	12

Table 4.3: Descriptive statistics of categorical wildfire and prescribed burn data.

	wildfire		prescribed fire	
metric	county	region	county	region
count	2041	2041	4123	4123
unique	97	8	51	7
top	Riverside	Central Valley	Siskiyou	Shasta Cascades
freq	178	443	624	1319

Figure 4.1 shows the distribution of wildland and prescribed fires over months and years. In Figure 4.1 we can see that wildfires peak from June to September, while prescribed fires peak from October to December. This result is expected since prescribed fires cannot be performed during peak wildfire season. With this in mind, comparing the amount of prescribed and wildland fire acres burned in the same month does not make sense for a predictor. Instead, using the engineered predictor for prescribed fires that incorporates different time lags is more suitable. The greatest amount of wildfire acres burned occurred in 2017, while 2019 saw the greatest amount of prescribed fires. Prescribed fires were far less frequent before 2018, with very few fires reported. Wildfires seem to be decreasing in frequency from 2017 onward.

Figure 4.2 shows the distribution of the natural log of wild and prescribed fire acres burned. Neither distribution is perfectly normal, though wildfire size has a right tail, indicating there are more large fires. Prescribed fires are generally smaller than wildfires with a more symmetric distribution. Interestingly, there are about 60 wildfire incidents and 40 prescribed fire incidents that report no acres burned.

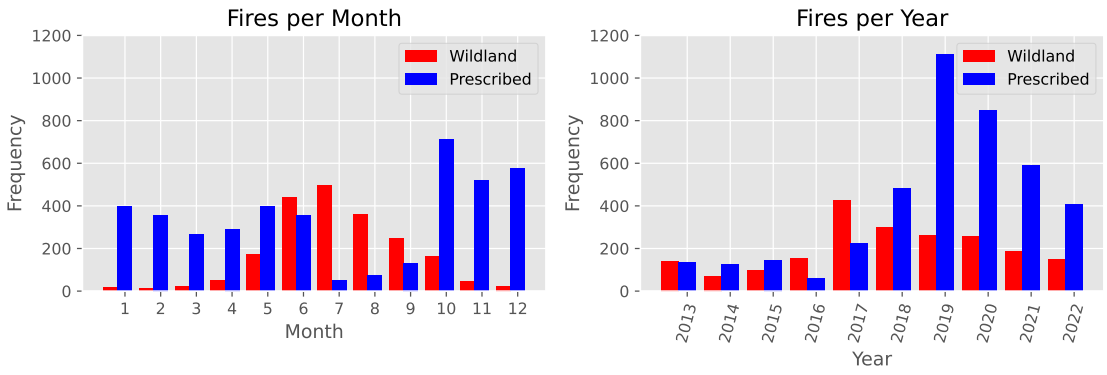


Figure 4.1: Distribution over years and months of wildfires and prescribed fires.

The total wild and prescribed fire acres burned per year from 2013-2022 is shown in Figure 4.3. Wildfires have far more acres burned than prescribed fires, with the most wildfire acres burned occurring in 2020. In contrast, prescribed burns had a lower amount of acres burned from 2013-2016, with the amount gradually increasing up to the peak in 2019. Additionally, 2019 had the most prescribed fire acres burned and also had a low number of wildfire acres burned. When combined with Figure 4.1, wildfires seem to be increasing in size in recent years, with 2021 and 2020 having the greatest amount of acres burned but fewer number of fire incidents in comparison to 2017 and 2018.

The maps of prescribed and wildland fires in Figures 4.4 and 4.5 do not show any clearly strong trends. Ideally, if prescribed burns are effective, there would be fewer and smaller wildland fires nearby in proceeding years. However, in 2019 and 2020 in Figure 4.5 there do appear to be fewer same-year fires close to prescribed

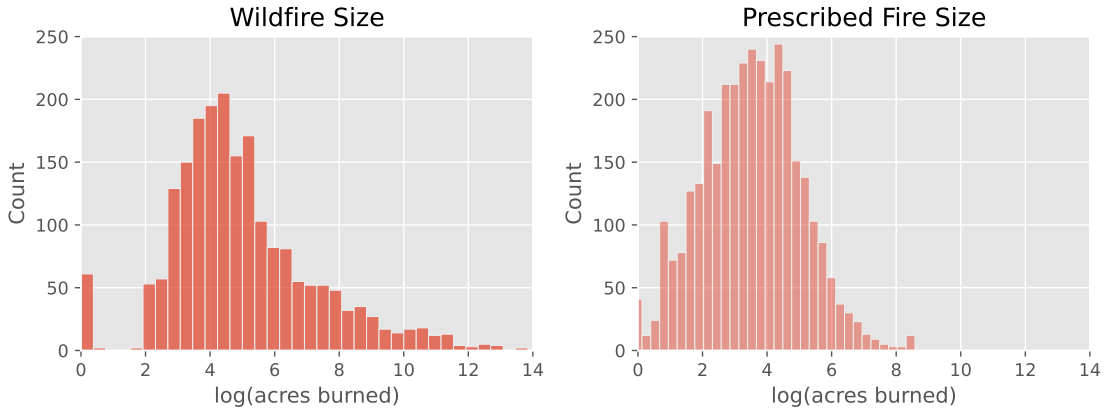


Figure 4.2: Distribution of wildfire and prescribed fire size.

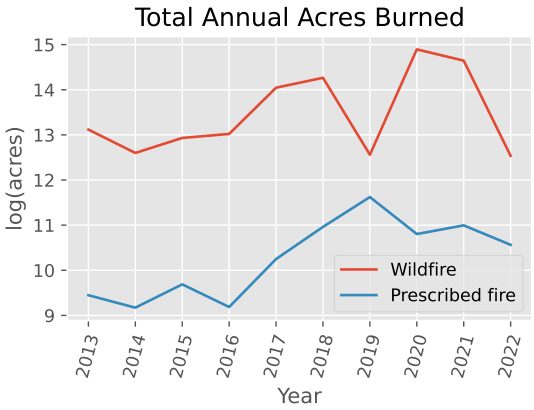


Figure 4.3: Total wild and prescribed fire acres burned per year.

burns.

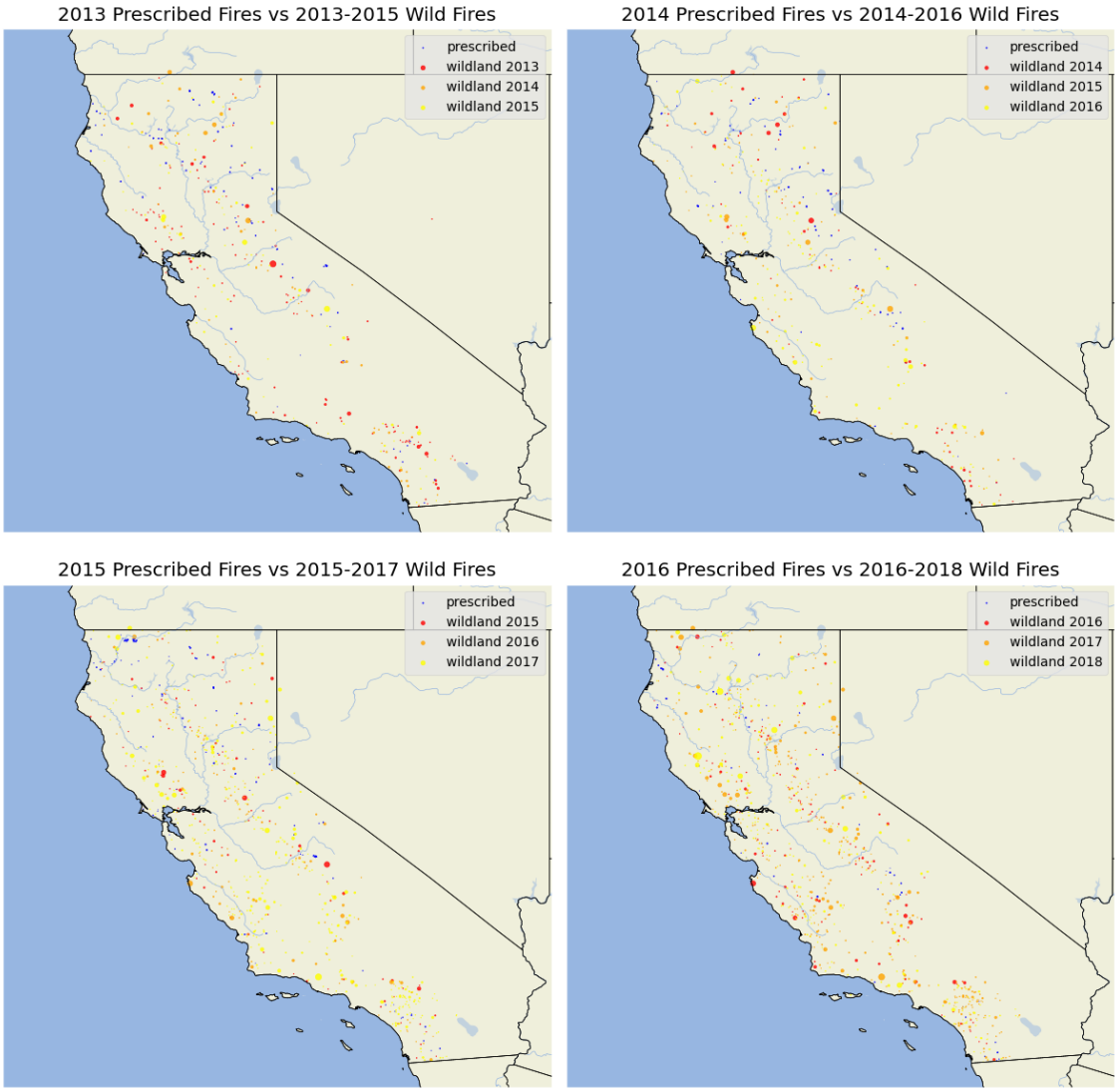


Figure 4.4: Location and size of prescribed fires and wildfires for 2013-2016.

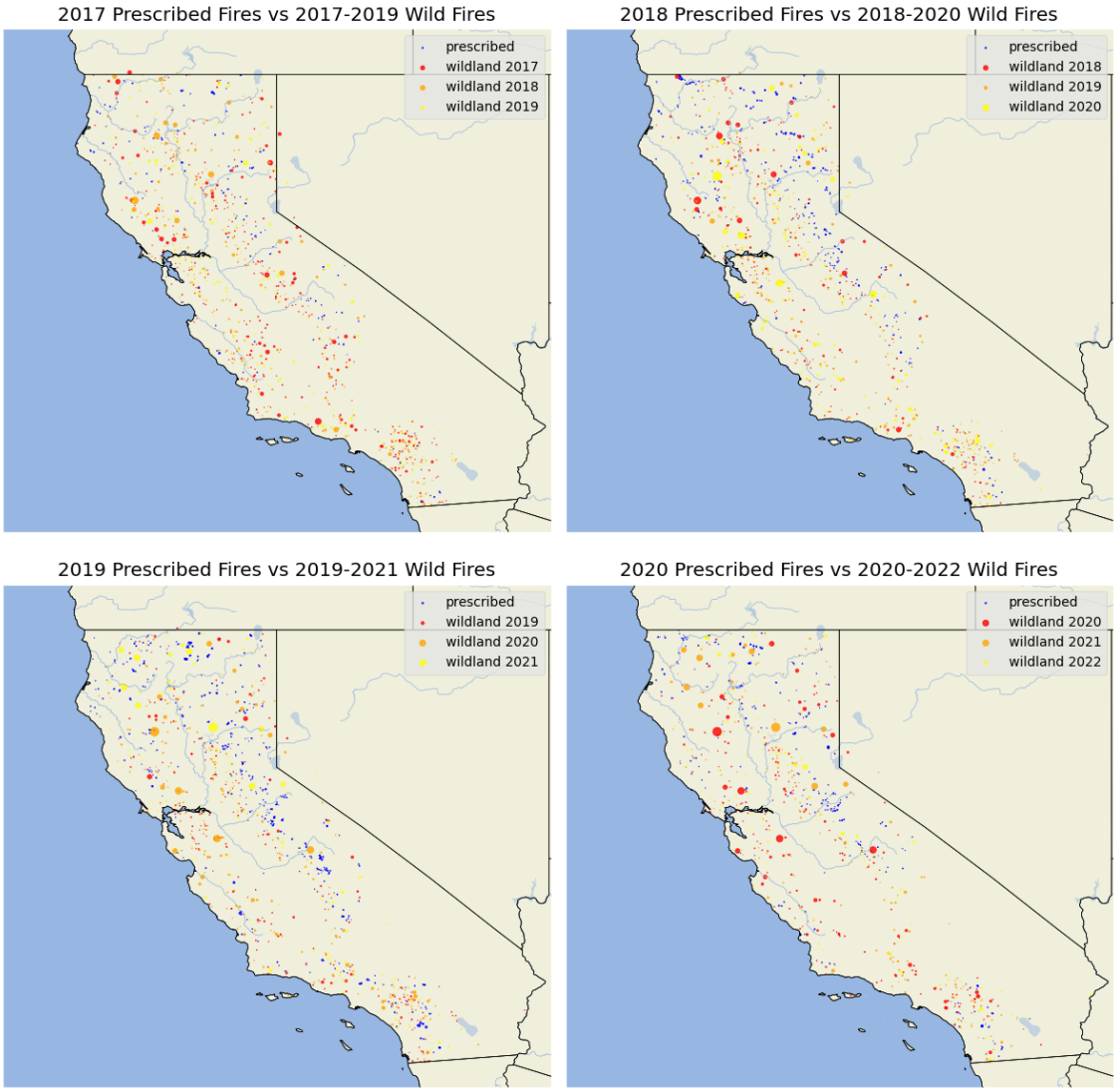


Figure 4.5: Location and size of prescribed fires and wildfires for 2017-2020.

4.2 Inferential Statistics

Inferential statistics methods are applied to the dataset to inform further analysis. In particular, the Mann-Whitney difference of median u-test and χ^2 independence test are both used to understand trends in the data.

4.2.1 Difference of Medians Tests

Difference of median tests can be used to test whether the median wildfire size is different between two different groups. In this case, prescribed fires and climate variables are binned to determine if they have an impact on wildfire size. Here a Mann-Whitney U-test (Wilcoxon rank-sum test) is used as a non-parametric difference of medians test on the monthly aggregated data[25]. The test statistic, U, is calculated according to Equation (4.1) and (4.2), where n_1 and n_2 are the number of observations in the first and second groups respectively and R_1 and R_2 are the sum of the pooled ranks in groups one and two respectively.

$$U = min(U_1, U_2) \tag{4.1}$$

$$U_1 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

$$U_2 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \tag{4.2}$$

If the probability of achieving a value as extreme as the test statistic is less that

0.05, then the result is considered statistically significant and there is evidence of a difference between the groups. The common language effect size (CLES) is used to analyze the strength of the effect in cases with significant differences. CLES is measured as the proportion of pairs where the condition of the tail is true [27]. Hence for a less-than tail, CLES is the proportion of pairs where the treatment group median is less than the control group median.

Climate's Influence on Wildfire Size

Four tests are performed for climate data, with one for each climate variable. Results of the tests are available in Table4.4 and box plots of the data is in Figure 4.6. For precipitation, months are grouped into those with an inch of rain or more and those with less than an inch of rain. The test results in a significant test statistic of $U = 11,277.5, (p < 0.0001)$, providing evidence that wildfires burn different amounts of acres when comparing months with and without at least an inch of precipitation. From the box plot in Figure4.6, it appears months with less than an inch of rain have more acres burned. When looking at temperature, months are grouped by having an average temperature less than sixty degrees Fahrenheit or greater than. Results of the test indicate a significant difference in wildfire acres burned ($U = 6,514.5, p = 0.0362$). Months with an average temperature greater than sixty degrees appear to have more wildfire acres burned in Figure4.6. Two drought indices are also examined, with months grouped by values less than and

greater than zero for both. Here a negative value indicates drought. For both PDSI and PZI, the tests found a significant difference in wildfire acres burned between the two groups, with test statistics $U = 12,767.5 (p < 0.0001)$ and $U = 12,034 (p < 0.0001)$ respectively. Hence, there is evidence that the amount of acres burned by wildfires is different between months with negative and positive drought indices. In the two box plots in Figure4.6, negative drought indices have larger amounts of acres burned.

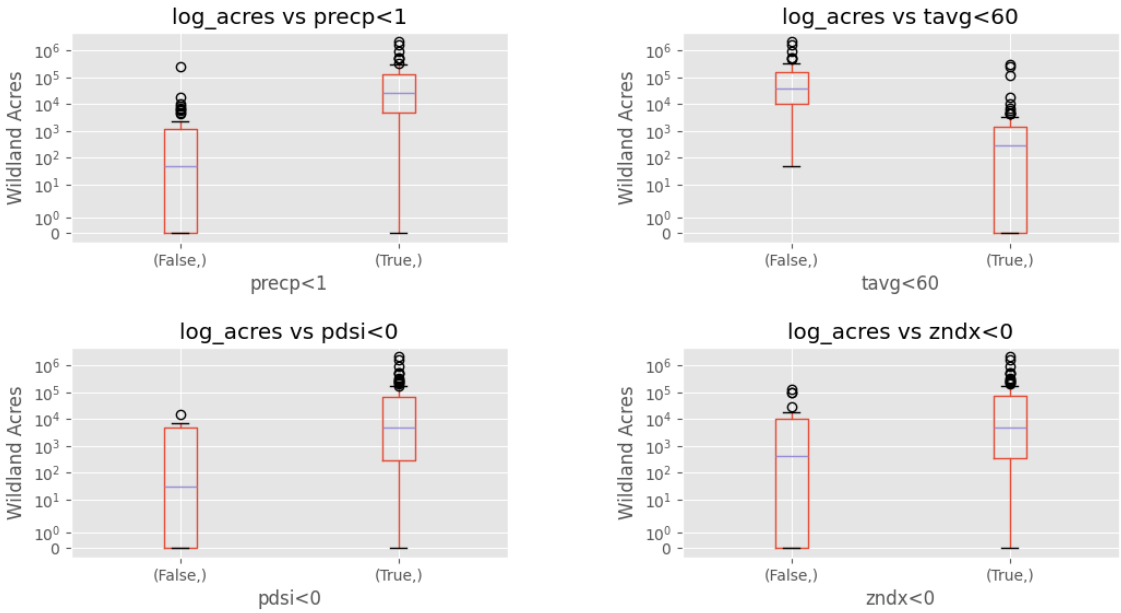


Figure 4.6: Boxplots for wildfire size according to climate variable conditions.

Table 4.4: Mann-Whitney U-test results for different climate variables.

condition	U-score	test type	p-val
precip < 1	11277.5	two-tailed	< 0.001
tavg < 60	6514.5	two-tailed	0.0036
pdsi < 0	12767.5	two-tailed	< 0.001
zndx < 0	12034.0	two-tailed	< 0.001

Prescribed Fire's Influence on Wildfire Size

Numerous Mann-Whitney U-tests are performed to analyze if prescribed fires occurring in varying spatial-temporal radii from wildfires result in smaller wildfires. For example, wildfires can be partitioned into those with a prescribed fire within 10 kilometers three months prior to ignition, and those without a prescribed fire within 10 kilometers in the past three months. Partitions for three, six, nine, twelve, twenty-four, and thirty-six month intervals are provided in Figures 4.7, 4.8, 4.9, 4.10, 4.11, and 4.12 respectively. From the box plots, the difference between the groups at different spatial resolutions is generally small. This difference between the sets of two groups' median wildfire size is compared using a one-sided U-test, where significance is tested for the tail which has smaller fires when a prescribed fire is present. Results from the U-tests are provided in Table 4.5, including the U-value, p-value, and common language effect size.

From the table, it is clear that prescribed fires occurring in any spatial distance from a wildfire in the previous three months results in smaller wildfires on average. In particular, prescribed fires within 10 kilometers have the greatest effect, the

57% of wildfires with a prior prescribed fire having a smaller size on average. At the six month interval, prescribed fires result in significantly smaller wildfires at the 10 ($p = 0.011$) and 50 kilometer (0.002) radii. A slightly smaller effect is seen at six months, with 54.5% of wildfires with a previous prescribed fire in 10 kilometers having a smaller size. Nine months before a wildfire, prescribed fires result in significantly smaller fires if they occur within 10 ($p = 0.016$), 50 ($p = 0.005$), or 100 kilometers ($p = 0.030$). At the nine month interval, prescribed fires within 100 kilometers have the greatest effect with 57.8% of fires treated with a prescribed burn resulting in fewer acres burned. For the tweleve, twenty-four, and thirty-six month temporal resolution the spatial range is constricted due to imbalanced data partitions for larger spatial ranges. Prescribed fires only result in significantly smaller wildfires when performed within 10 ($p = 0.040$) and 50 kilometers ($p = 0.0002$) twelve months prior to a wildfire. At twenty-four months before a wildfire, prescribed fires result in significantly fewer acres burned when they occur within 10 ($p = 0.050$), 25 ($p = 0.048$), and 50 kilometers ($p = 0.012$). At the thirty-six month interval, prescribed fires are only effective at 25 ($p = 0.017$) and 50 kilometer ($p = 0.007$) distances. Prescribed fires are most effective at the 50 kilometer range when performed at twelve, twenty-four, and thirty-six months before a fire, with 56.5%, 55.4%, and 57% of treated wildfires having fewer acres burned than non-treated fires respectively.

From this analysis, it is clear that prescribed fires can have a significant impact

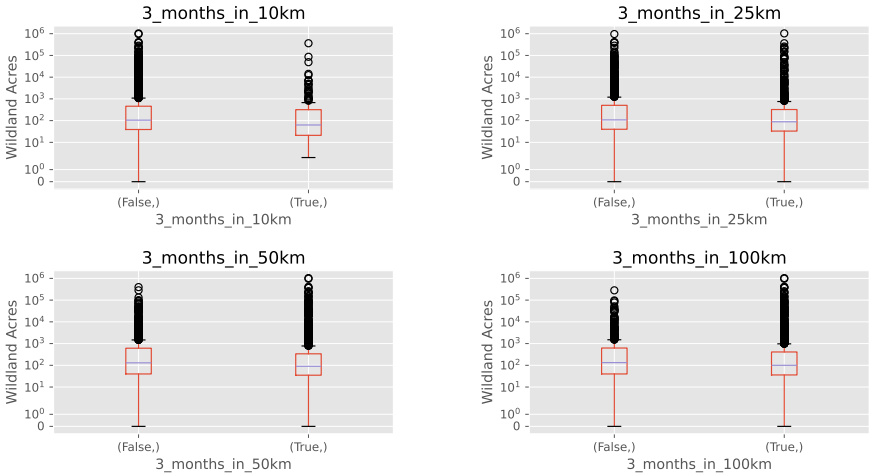


Figure 4.7: Distribution of wildfires with prescribed fires in varies spatial radii three months prior to ignition.

on the size of wildfires depending on when and where they occur. The largest impact of prescribed burns varies on the temporal scale; however, the 50 kilometer range consistently results in significant effects. While this test is useful, the U-test fails to account for confounding variables, such as climate conditions, and thus should be considered carefully.

4.2.2 Independence Tests

The χ^2 independence test can be used to determine if year and region are independent when considering fire frequency. In this case, a null hypothesis would consider fire frequency the same regardless of year and region, whereas the alternative hypothesis would be that fire frequency varies in year and region. That is, the amount

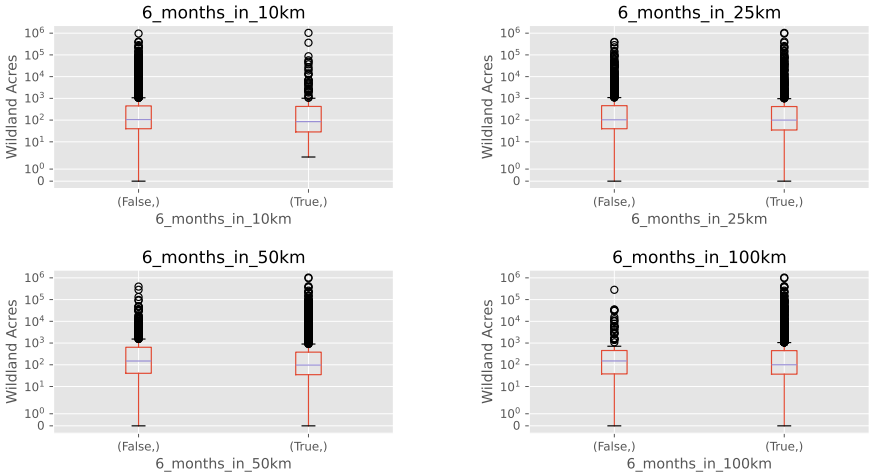


Figure 4.8: Distribution of wildfires with prescribed fires in varies spatial radii six months prior to ignition.

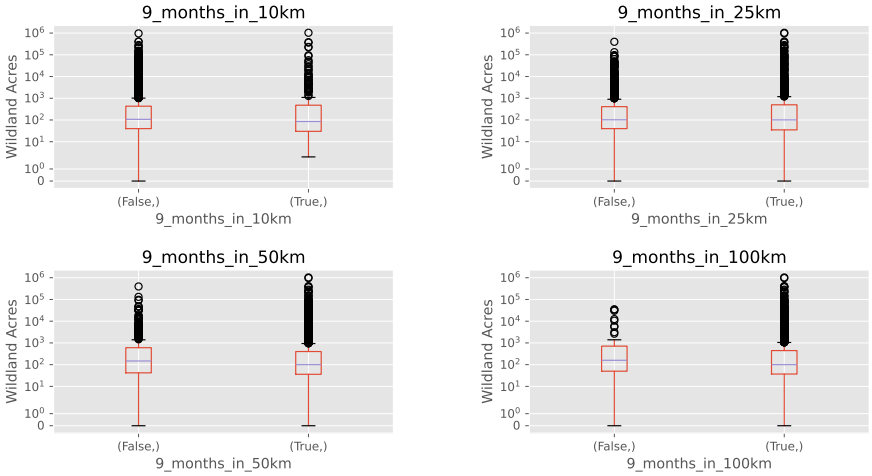


Figure 4.9: Distribution of wildfires with prescribed fires in varies spatial radii nine months prior to ignition.

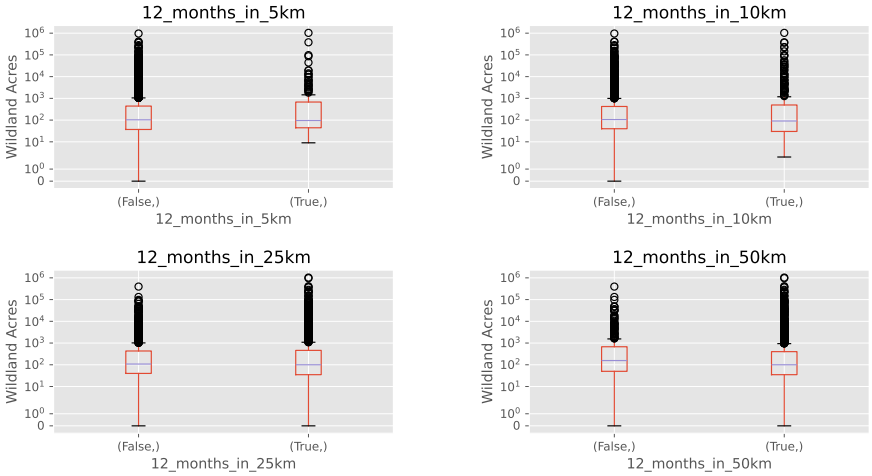


Figure 4.10: Distribution of wildfires with prescribed fires in varies spatial radii twelve months prior to ignition.

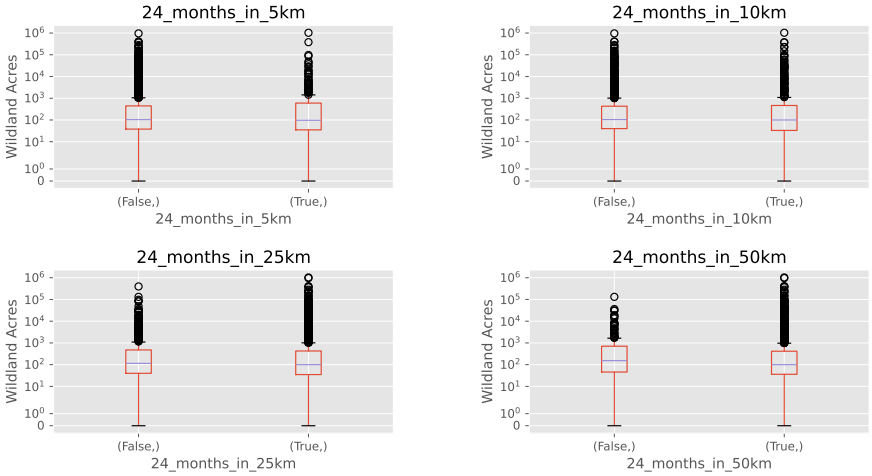


Figure 4.11: Distribution of wildfires with prescribed fires in varies spatial radii twenty-four months prior to ignition.

Table 4.5: Mann-Whitney U-test results with common language effect size (CLES).

Month	Spatial Radius	U-value	Tail	P-val	CLES
3	10km	116580.5	Less	**0.002**	0.570
	25km	398966.5	Less	**0.015**	0.531
	50km	47.0224.5	Less	**0.0005**	0.542
	100km	324365	Less	**0.054**	0.527
6	10km	197234	Less	**0.011**	0.545
	25km	505585.5	Less	0.345	0.505
	50km	369592	Less	**0.002**	0.542
	100km	134623.5	Less	0.081	0.534
9	10km	253330.5	Less	**0.016**	0.538
	25km	519750.0	Less	0.475	0.501
	50km	284212.5	Less	**0.005**	0.543
	100km	41153	Less	**0.030**	0.578
12	5 km	136863	Less	0.618	0.492
	10km	298695	Less	**0.040**	0.529
	25km	500068.5	Less	0.195	0.511
	50km	21464	Less	**0.0002**	0.566
24	5 km	191306	Less	0.414	0.505
	10km	385478	Less	**0.050**	0.524
	25km	434609.5	Less	**0.048**	0.523
	50km	133410	Less	**0.012**	0.554
36	5 km	213258	Less	0.273	0.512
	10km	434751.5	Less	0.086	0.519
	25km	385354.5	Less	**0.017**	0.531
	50km	90589.5	Less	**0.007**	0.570

of fires in one region depends on the year. The hypothesis test is conducted by calculating a test statistic based on the observed category counts, n_i, and expected counts given independence, μ_i, defined in Equation (4.3). This statistic follows a χ^2 distribution, which is used to determine the probability of obtaining the value of

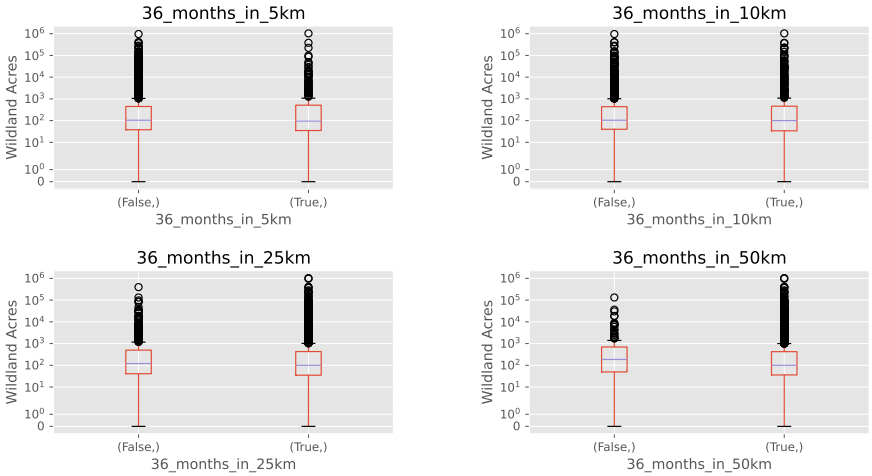


Figure 4.12: Distribution of wildfires with prescribed fires in varies spatial radii thirty-six months prior to ignition.

the test statistic.

$$X^2 = \sum_{i=1}^{k} \frac{(n_i - \mu_i)^2}{\mu_i} \tag{4.3}$$

The Chi-squared independence test is performed for both wild and prescribed fires, with the results of the statistical test displayed in Table 4.6. For both prescribed and wild fires the test results in a significant statistic at the 99% significance level, rejecting the null hypothesis that fire frequency is the same regardless of year and region. To better understand how fire frequency varies across years and regions, Figures 4.13 and 4.13 show the expected counts, observed counts, and standardized residuals for wildfires and prescribed fires respectively. Any standardized residual

with a magnitude greater than three is considered a large deviation from the expected value.

From Figure 4.13, we can see that there were more wildfires than expected in the Shasta Cascades in both 2018 and 2019, while 2016 and 2022 saw fewer wildfires than expected in the same region. The Sierras also had high variation, with fewer fires than expected in 2015 and 2019 and more fires than expected in 2020 and 2021. The Central Valley only experienced excessive fires in 2019, with fewer fires in 2021 and 2022. For the Central Coast, 2017 and 2022 saw more fires than expected, while 2020 saw fewer. The Desert, South Coast, and North Coast regions had less variation overall, With only the South Coast having excessive fires in 2021.

Similar to wildfires, Figure 4.14 shows the Desert and South Coast regions have less severe deviations from the expected distribution of prescribed fires. Interestingly, the North Coast region has more prescribed fires than anticipated prior to 2018, then drops to slightly fewer prescribed fires than anticipated from 2018-2022. The Central Valley saw more prescribed burns in 2019, followed by fewer in 2021 and 2022. The Shasta Cascades also saw an uptick in prescribed fires in 2018 and 2019, followed by a decrease in 2021 and 2022. In the Sierras, fewer prescribed fires than expected occurred in 2015 and 2019, with more prescribed fires in 2020 and 2021. There is also variation in the Central Coast region, with more prescribed fires than expected in 2017 and 2022 and fewer in 2020. Overall, the test shows region and year are not independent for both wild and prescribed fires and indicates an

interaction where regions have more or less fires than expected for different years.

Table 4.6: χ^2 Independence test statistic results for region and year.

Fire Type	χ^2	degrees of freedom	p-val
Wild Fire	126.836	63	< 0.001
Prescribed Fire	382.960	54	< 0.001

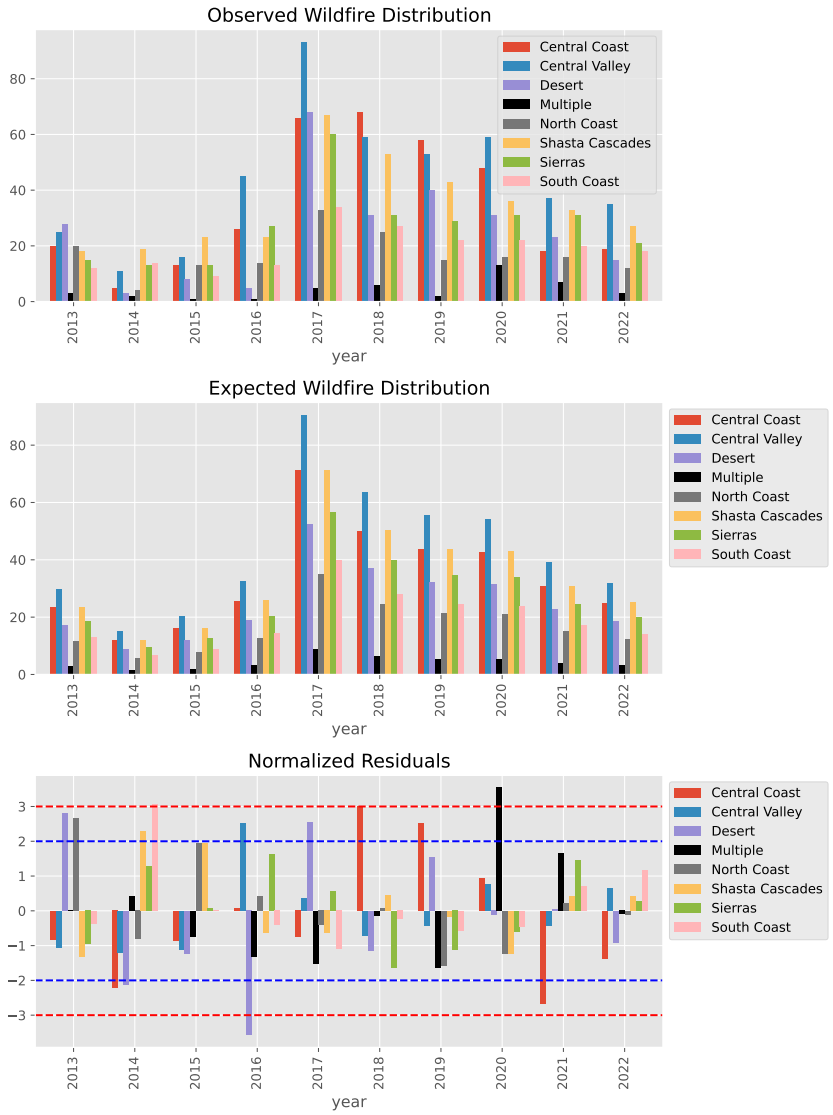


Figure 4.13: Observed, expected, and residuals for wildfire frequency between regions and years.

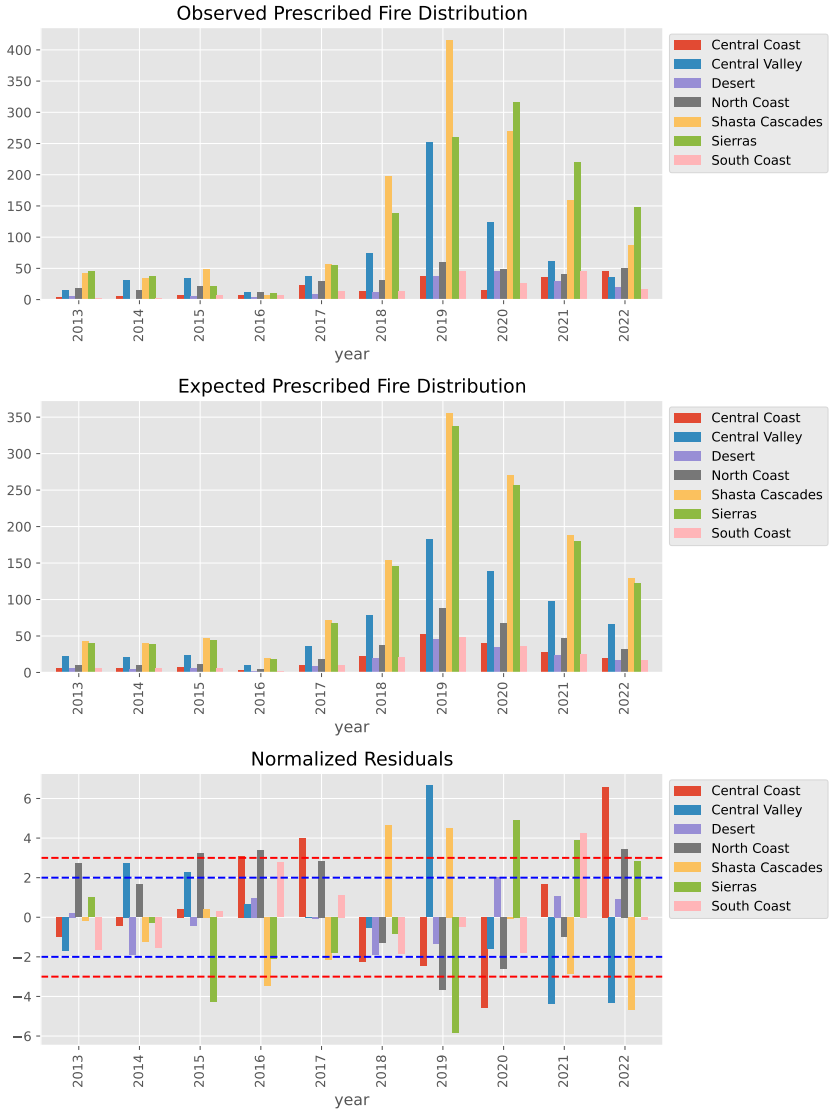


Figure 4.14: Observed, expected, and residuals for prescribed fire frequency between regions and years.

Chapter 5

Time Series Analysis

We can treat the amount of wildfire acres burned each month, as well as predictors, as a time series to examine how trends vary over time. In particular, time series graphs, correlation, and cross-correlation are used to visualize how variables fluctuate and their pairwise relations.

5.1 Variable Time Series

Time series graphs of monthly wildfires acres burned, prescribed fire acres burned, and climate variables are displayed in Figure 5.1. Precipitation and temperature variables cycle between highs and lows throughout each year, with some years showing low precipitation. The four drought index variables fluctuate less over months and instead have greater variation over years, besides the Z-index. When comparing graphs, it is clear the drought indices all follow the same trends, similar to the same trends viewed in the temperature data. Unsurprisingly precipitation peaks

appear to correspond to drought index peaks, suggesting drought is closely coupled to precipitation.

Consistent with Figure 4.3, in Figure 5.1 wildfire acres burned tend to spike periodically throughout the year, with only summer months seeing large values. The peaks of the wildfire acres burned appear to grow over time, especially when comparing 2013-2016 with 2017-2021. Prescribed fires have a reverse pattern, with fewer acres burned in summer months and peaks during other months. This trend is also visible in Figure5.2, which shows the prescribed and wildfire acres burned time series on the same axis. From Figure5.2, wildfire acres burned and prescribed fire acres burned can be more clearly compared. On the logarithmic scale, there are annual trends where some months have more prescribed fire acres than wildfire acres, likely corresponding to winter months. In both the graphs in Figure5.2 the wildfire acres burned peaks can be seen occurring regularly each year with the maximum values much greater than the prescribed fire peaks. It appears in 2019 there was a high amount of prescribed fire acres and a low amount of wildfire acres, which could indicate effective prescribed burns. The highest amount of prescribed fire acres appears from 2017 onward, perhaps indicating policy change encouraging prescribed burns. Since the effect of prescribed fires appears in the months following a burn, additional variables for the total sum of prescribed acres burned over the previous three-, six-, nine-, and twelve-month intervals are also calculated. In Figure 5.1, these aggregate prescribed burn time series result in a gradually smoother curve

than the raw monthly time series.

5.2 Correlation

To better quantify the pairwise relationship between variables, a correlation matrix is shown in Figure 5.3. Each cell in the correlation matrix corresponds to the correlation between the row, x, and column, y variables, as defined by R in Equation (5.1).

$$R = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{5.1}$$

The variable of interest, the log of wildfire acres burned, is in the top row of the matrix with its correlation to each predictor shown in the corresponding columns. Of note, wildfire acres burned have the weakest correlation with prescribed fire acres burned, even when considering prescribed fires over different time periods. However, the prescribed fire acres in a three-month period do have a negative association, which is the trend direction that demonstrates prescribed fires are related to reduced wildfire size.

Wildfire acres are most strongly associated with temperature, where temperature increases are directly associated with increases in wildfire acres burned. Precipitation has a strong negative correlation with wildfires, while the drought indices have a weak negative correlation, thus decreased precipitation and severe drought are

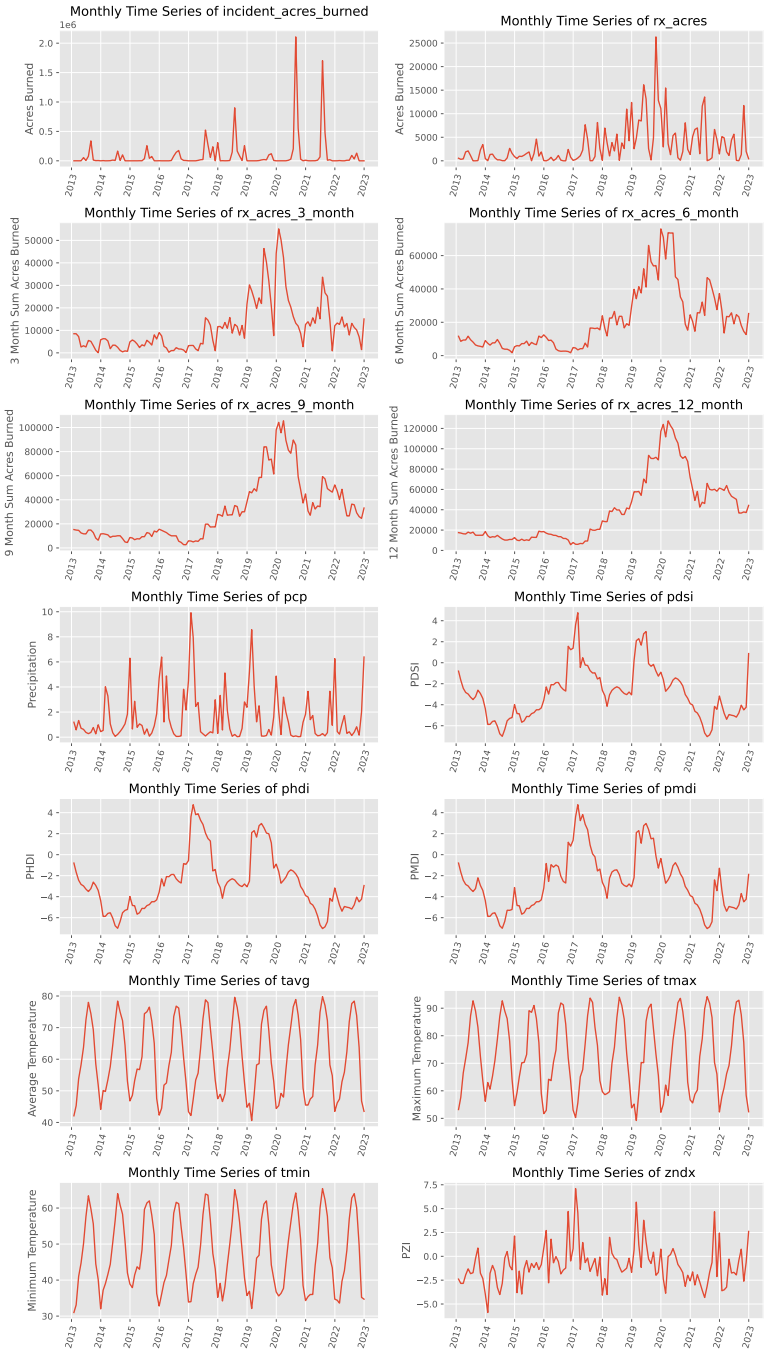


Figure 5.1: Time series of predictors.

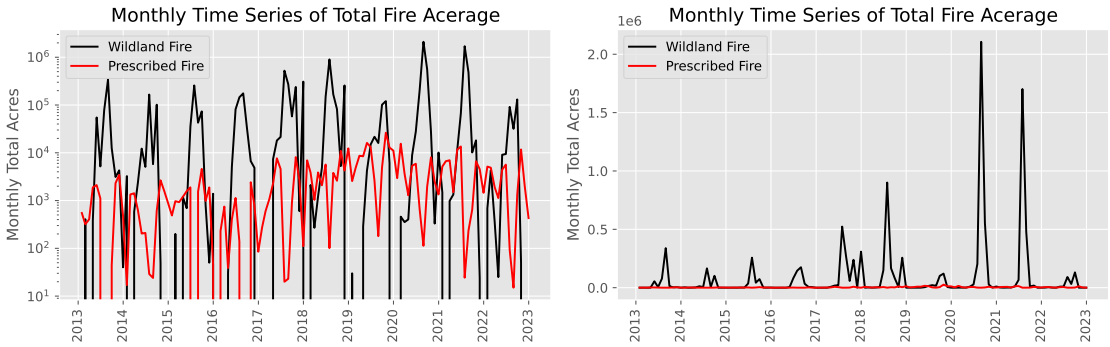


Figure 5.2: Wildfire versus prescribed fire time series.

associated with greater wildfire acres burned.

A key observation is the correlation between the predictors, where the temperature variables, drought indices, and prescribed fire aggregates are all strongly positively correlated within their own groups. This signals that multicollinearity may be a problem for regression models, and instead only one variable of each group should be chosen for models. Overall, Figure 5.3 shows wildfires are more strongly related to climate variables rather than prescribed fires, indicating changing climate conditions may influence wildfires more so than prescribed fires.

5.3 Cross-correlation

A cross-correlation analysis is performed to identify the temporal lag that produces the strongest relationship between wildfire acres burned and the predictors. For a cross-correlation of lag d, the correlation coefficient is calculated by shifting the time

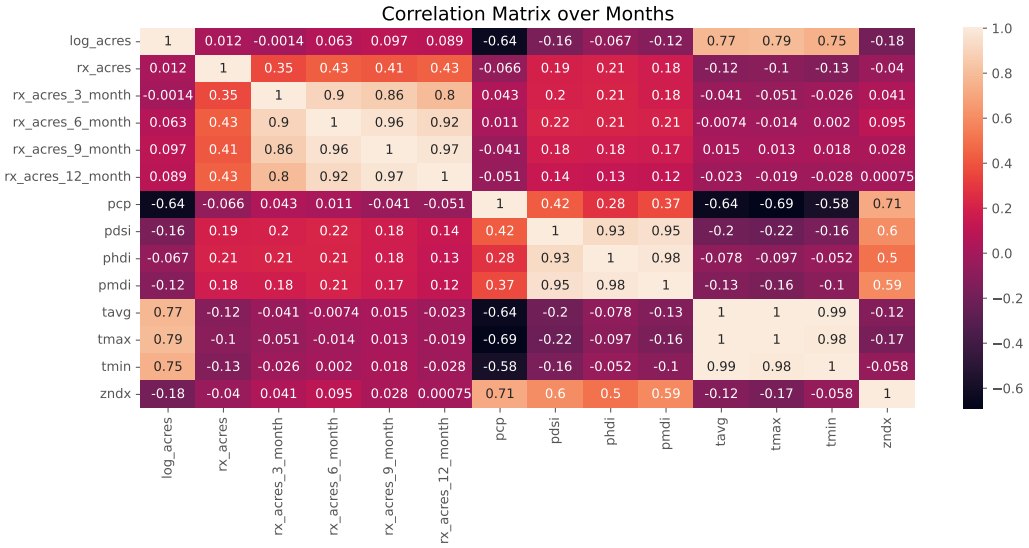


Figure 5.3: Correlation matrix for predictors.

series of a variable by d time steps, as defined in Equation (5.2).

$$R_d = \frac{\sum_{i=1}^{n-d}(x_{i+d} - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n-d}(x_{i+d} - \bar{x})^2}\sqrt{\sum_{i=1}^{n-d}(y_i - \bar{y})^2}} \tag{5.2}$$

Both positive and negative correlations are considered over a span of 24 months, with results in Figure 5.4 and Table 5.1. Positive correlations indicate variables that are associated with increased wildfire size, while negative correlations are associated with decreased wildfire size. Thus when considering wildfire mitigation strategies, negative correlations are insightful.

In Table 5.1, the strongest negative correlations all occur within the 0-6 month time frame. This trend suggests that conditions in the previous year are most

important for wildfire mitigation. Prescribed fires have the strongest impact at the 2-month lag, yet the association is still weak with $R = -0.077$. With this in mind, prescribed fires should be performed as close to fire season as possible to be most effective at reducing wildfire size.

Temperature has the strongest relationship to wildfire acres burned. When considering the temperature in the same month as the recorded wildfire acres burned (e.g., lag=0), the maximum temperature has a strong positive correlation to acres burned with $R = 0.788$. This result implies that extreme temperature events in the current month are most closely related to the number of wildfire acres burned in a given month. Temperature also has the strongest negative relationship to wildfire acres burned when considering 6 months before the acres burned, with $R = -0.703$. This relationship is likely due to peak fire season occurring in summer when temperatures are high, while winter is approximately 6 months before fire season with low temperatures.

Precipitation also has a strong relationship to wildfire acres burned, with a strong negative correlation of $R = -0.636$ at a lag of 0. Hence, more precipitation is associated with fewer acres burned in a given month, indicating precipitation could be a protective factor. This trend could be explained by most precipitation occurring outside of wildfire season, though some regions do experience mild summer storms.

The drought indices (phdi, pmdi,pdsi,zndx) all have the strongest negative correlation to wildfire acres burned when considering the same month time scale (lag=0).

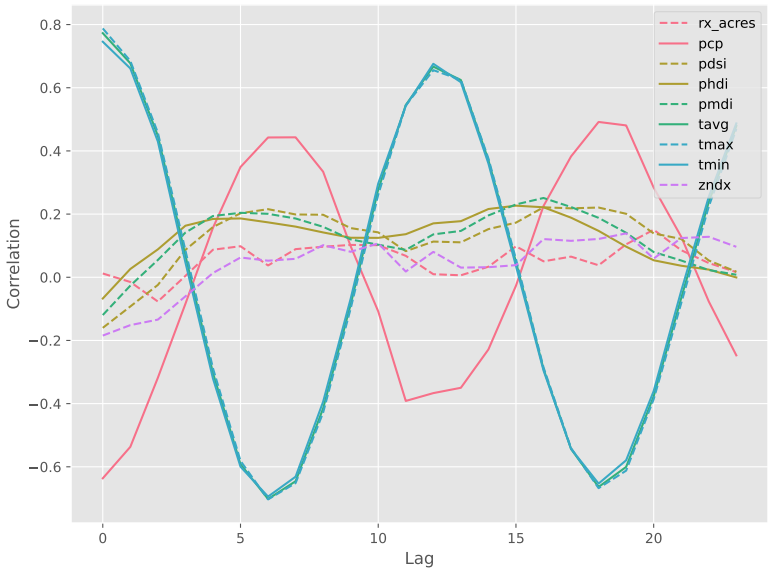


Figure 5.4: Cross correlation of predictors and wildfire acres burned.

The Palmer Z-index has the strongest negative correlation of the drought indices at $R = -0.185$. While a weaker association, this trend suggests that months with severe drought have more wildfire acres burned.

5.4 Discussion

The time series analysis in this section provides an overview of longitudinal trends in wildfires, prescribed fires, and climate variables. This analysis examined which variables are most strongly related to wildfire acres burned and identified the temporal period predictors had the strongest association. A recurring theme in this analysis

Table 5.1: Cross correlation results.

predictor	best pos lag	max R	best neg lag	min R
rx acres	20	0.150	2	-0.077
pcp	18	0.492	0	-0.636
pdsi	16	0.222	0	-0.160
phdi	15	0.227	0	-0.067
pmdi	16	0.251	0	-0.120
tavg	0	0.773	6	-0.702
tmax	0	**0.788**	6	**-0.703**
tmin	0	0.746	6	-0.694
zndx	19	0.139	0	-0.185

is the impact of climate variables compared to prescribed fires. When considering the data over all of California in monthly totals, prescribed fires have a minimal impact on wildfire acres burned, while climate variables, especially temperature and precipitation, have a large association with wildfire acres burned. One limitation of this approach is that it fails to take into account differences within a region. Certain regions may have a stronger effect of prescribed burns than seen when examining the entire state. This time series also fails to consider the spatial distance between wildfires and prescribed fires, which may also affect the relationship.

Chapter 6

Regression Analysis

By modeling wildfire size in acres as a regression problem, the most important predictors of wildfire size can be identified. In this section, each wildfire is treated as a data point where the target variable, acres burned, is modeled using regression. A linear regression, mixed linear effects model, support vector machine, k-nearest neighbors, random forest, and boosting model are all tested.

6.1 Preprocessing

Wildfires with no acres burned are removed from the analysis, resulting in a final sample size of 1969 wildfires. As seen in the Figure 5.3, many of the predictors are correlated with each other. Before using linear models, correlated predictors are removed using the variation inflation factor (VIF). VIF is calculated for the entire set of predictors except for the prescribed fire variables, with initial scores in Table 6.1. Next, variables with the highest VIF are removed sequentially until

all variables have a VIF around 10, resulting in the scores in Table 6.2. Since the climate variables had the most multicollinearity, the maximum temperature, precipitation and maximum windspeed were chosen for their ease of interpretation and independence. Before model training, each variable is standardized using its sample mean, $\bar{x}_i$ and standard deviation, s_i, in the equation below:

$$x^*_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i}$$

The sample mean and standard deviation in the scaling are calculated using the training set, then used to transform the test set and validation sets.

Table 6.1: Variation inflation factor for potential predictors.

feature	VIF
longitude	606.215
latitude	375.176
tmax	110.998
pcp	2.354
zndx	1.795
max wind	15.290
log rx 3 months 50km	4.474
log rx 12 months 50km	12.973
log rx 6 months 50km	11.687

Table 6.2: Variation inflation factor for subset of predictors.

feature	VIF
pcp	1.277
log_RX_3_mo_50km	2.121
tmax	13.582
max_wind	12.686

6.2 Model Selection

All models are trained and fine-tuned using 5-fold cross-validation, with a test set used for final model evaluation. The data is split with 80% for training and cross-validation and 20% for testing. The training and cross-validation data is split again with 5-fold cross-validation. The cross-validation is primarily used to select the best parameters and predictors for non-linear models, while the test set is used to estimate how well the model generalizes. Once the best parameters for a model are selected using cross-validation, the model is refitted with the full training and validation dataset (80%) and evaluated on the test set. Model performance is measured using mean squared error (MSE), defined in Equation 6.1.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2 \tag{6.1}$$

Six different models are selected for training and evaluation, including linear regression, mixed effect linear model, support vector machine regression, nearest neighbors regression, random forest, and adaboost with decision tree estimators. The linear models are useful for inference on the effect of each predictor on the

target and provide easy interpretation for how changes in predictors impact the target. All linear models are fit with the full selected subset of predictors (drought index, maximum temperature, precipitation, maximum wind, and prescribed burn variables), then insignificant predictors are removed. The remaining models are fit with the full subset of predictors plus additional prescribed fire variables, maximum temperature, latitude, and longitude, with the best parameters selected using cross-validation.

6.2.1 Linear Regression

Linear regression models a response variable Y, in this case the natural logarithm of wildfire acres burned, as a linear combination of p predictors, given in Equation (6.2).

$$Y = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p + \epsilon \tag{6.2}$$

A linear regression model estimates the response variable, $\hat{Y}$, by estimating the coefficients of the predictors, $\hat{\beta}_j$, using the ordinary least squares (OLS) method. OLS finds the set of $\hat{\beta}_j$ such that the squared error (Equation (6.3)) is minimized.

$$L = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \tag{6.3}$$

While simple in nature, linear regression provides a clear interpretation of the effects of each predictor on the response. In this section, linear regression is used

to identify which predictors are related to wildfire size and the direction of the association. Linear regression is applied to individual regions separately to see how predictors vary, as well as in a general model.

Regional Modeling

A separate linear regression model can be fit to each region individually to identify which predictors are relevant in each region. Here the predictors of interest are precipitation, maximum wind speed, maximum temperature, and prescribed fire acres within three months prior and 50 kilometers. Table 6.3 shows the performance of the regional models, while Figure 6.1 shows the coefficients for the predictors in each region.

In Table 6.3 it appears that performance across regions varies, with the best model on fires in multiple regions having an $R^2 = 0.937$ and the worst model in the North Coast region having an $R^2 = 0.863$. Interestingly, the multiple region model has the highest train and test error, indicating fires in multiple regions have larger variance in size. The Central Valley model performs the best on the test set with $MSE = 2.978$, while the Central Coast model performs the best on the train set with $MSE = 2.501$.

In Figure 6.1 coefficients with significant < 0.01 are marked with three stars, < 0.05 marked with two stars, and < 0.1 marked with one star. Predictor coefficients are mostly negative. Hence, increased precipitation, greater prescribed burn acres,

and higher temperature are generally associated with fewer acres burned.

The direction of the relationships between the predictors and wildfire acres burned is mostly consistent across regions, with more precipitation only significantly related to greater acres burned in the desert and more prescribed fires related to greater acres burned in fires spanning multiple regions. The magnitude of the coefficients does vary across regions, with fires in multiple regions and the Sierras having larger coefficients for maximum wind speed, and fires in multiple regions having a larger magnitude coefficient for precipitation.

Precipitation and maximum temperature tend to have larger coefficients than the prescribed fire variables, indicating climate may play a greater role in wildfire size than prescribed burns in certain regions. In sum, prescribed fires are related to decreased wildfire acres burned in certain regions only, even when controlling for precipitation, temperature, and wind speed.

Table 6.3: Performance for separate models for each region.

Region	Train MSE	Test MSE	R^2
Sierras	3.771	6.146	0.880
Desert	2.508	4.249	0.885
Central Valley	2.877	**2.978**	0.9045
North Coast	3.537	3.005	0.863
Central Coast	2.501	3.471	0.902
Multiple	6.472	11.423	**0.937**
South Coast	3.199	3.885	0.904
Shasta Cascades	4.601	6.462	0.853

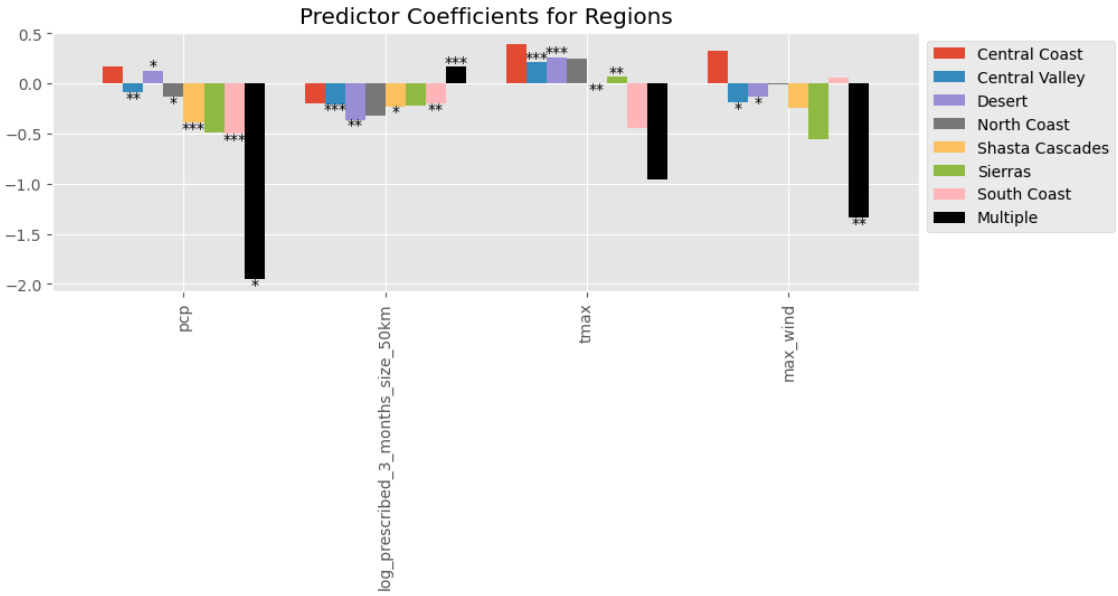


Figure 6.1: Predictor coefficients for each region.

General Model

To understand how the number of wildfire acres burned is affected by different predictors across the state, one general model is fit. The predictors in this model are the same as the regional models (precipitation, maximum wind speed, maximum temperature, prescribed fire acres within 3 months prior and 50 kms) plus a categorical variable for region and interactions between region and the other variables. Only a subset of regions are included in the model after fitting a model with all regions and grouping non-significant regions into one category. This results in the following regions: Sierras, Shasta Cascades, Central Coast, South Coast, and Multiple.

Regression results with coefficients, p-values, and confidence intervals are avail-

Table 6.4: Linear regression results.

	Estimate	Std Error	t	P($>\|t\|$)	95% CI
***constant	4.2186	0.681	6.198	0.000	[2.884, 5.554]
pcp	-0.0889	0.082	-1.078	0.281	[-0.251, 0.073]
***log_RX_3_mo_50km	-0.3456	0.074	-4.678	0.000	[-0.490, -0.201]
*tmax	0.1349	0.081	1.660	0.097	[-0.024, 0.294]
max_wind	-0.1101	0.071	-1.555	0.120	[-0.249 , 0.029]
Sierras_X_tmax	-0.1806	0.561	-0.322	0.748	[-1.281, 0.920]
**Sierras_X_pcp	-0.1651	0.076	-2.181	0.029	[-0.314, -0.017]
Sierras_X_log_RX_3_mo_50km	0.0658	0.093	0.705	0.481	[-0.117, 0.249]
***Sierras_X_max_wind	-0.6907	0.216	-3.192	0.001	[-1.115, -0.266]
***South Coast_X_tmax	-1.7193	0.587	-2.931	0.003	[-2.870, -0.569]
**South Coast_X_pcp	-0.1619	0.065	-2.505	0.012	[-0.289, -0.035]
South Coast_X_log_RX_3_mo_50km	0.0393	0.063	0.622	0.534	[-0.085, 0.163]
South Coast_X_max_wind	0.1850	0.186	0.993	0.321	[-0.180, 0.550]
*Central Coast_X_tmax	1.2475	0.712	1.753	0.080	[-0.148, 2.643]
Central Coast_X_pcp	0.1116	0.076	1.469	0.142	[-0.037, 0.261]
Central Coast_X_log_RX_3_mo_50km	0.0572	0.068	0.838	0.402	[-0.077, 0.191]
***Central Coast_X_max_wind	0.6598	0.205	3.213	0.001	[0.257, 1.062]
Shasta Cascades_X_tmax	-0.4328	0.498	-0.869	0.385	[-1.409, 0.544]
*Shasta Cascades_X_pcp	-0.1364	0.082	-1.666	0.096	[-0.297, 0.024]
Shasta Cascades_X_log_RX_3_mo_50km	0.0747	0.082	0.911	0.362	[-0.086, 0.235]
Shasta Cascades_X_max_wind	-0.2716	0.232	-1.168	0.243	[-0.728, 0.184]
*Multiple_X_tmax	-1.6702	1.025	-1.629	0.104	[-3.682, 0.341]
***Multiple_X_pcp	-0.2968	0.106	-2.803	0.005	[-0.504, -0.089]
Multiple_X_log_RX_3_mo_50km	0.1264	0.092	1.367	0.172	[-0.055, 0.308]
***Multiple_X_max_wind	-0.6906	0.205	-3.362	0.001	[-1.094, -0.288]
*Sierras	3.0901	1.805	1.712	0.087	[-0.450, 6.630]
***South Coast	5.9419	2.228	2.667	0.008	[1.572, 10.312]
***Central Coast	-5.2917	2.019	-2.620	0.009	[-9.253, -1.331]
Shasta Cascades	2.2946	1.482	1.549	0.122	[-0.612, 5.201]
***Multiple	21.0104	7.655	2.745	0.006	[5.995, 36.026]
F-stat	405.7				
F p-val	< 0.001				
Adj. R^2	0.885				

able in Table 6.4, with significant coefficients marked with stars. The model is a good fit for the data and explains 88.5% of the variation in wildfire acres burned with an adjusted $R^2 = 0.885$. The model fit is also significant, with $F = 405.7 (p < 0.001)$.

To further demonstrate the quality of the model, the fitted model is compared to a null model with just a constant using an Analysis of Variance (ANOVA) test, with results in Table 6.5. The ANOVA finds the fitted model is significantly different from the null model, with $F = 12.756 (p < 0.001)$.

Table 6.5: ANOVA for linear model vs null model.

	df residual	SSR	df diff	SS diff	F	Pr(>F)
0	1574	6629.645	0.0	-	-	-
1	1545	5348.951	29.0	1280.694	12.756	< 0.001

In Table 6.4, the acres burned in prescribed fires within 3 months and 50 km is significant on its own, though none of its interactions with region are significant. The coefficient for the prescribed burns variable is negative, indicating fires with greater prescribed acres burned tend to have fewer wildfire acres burned. Additionally, the coefficient magnitude of the prescribed fires in 3 months and 50 km variable is the largest among the climate and prescribed fire variables.

Another observation in Table 6.4 is that both precipitation and maximum wind speed are insignificant on their own; however, these variables have significant interactions with the region variables. This entails the impact of precipitation and wind may vary depending on the region. Both wind speed and precipitation tend to have negative coefficients when significant, though wind speed has a positive coefficient for the Hence in the Central Coast, wildfires experiencing a high wind speed tend to be larger than those with lower wind speeds. The interaction between precipitation and the Central Coast is not significant, while the interaction is significant with

the other four regions (Sierras, South Coast, Shasta, Multiple). This means that wildfires in the Sierras and Multiple regions have a greater reduction in acres burned on average with increases in precipitation when compared to other regions. That is, precipitation is a stronger reducer of wildfire size in fires in the Sierras and spanning Multiple regions. Temperature is a significant predictor in the model, both on its own and in interactions with South Coast and Central Coast regions. Increases in temperature are significantly associated with increases in wildfire size, except in the South Coast region.

The effect of the four primary predictors of interest can be examined through plots, with Figures 6.2, 6.3, 6.4, and 6.5 showing the residuals, true versus fitted values, partial regression plot, and component and component-plus-residual (CCPR) plot for each of the predictors. The true versus fitted value graphs have a confidence interval on each point, which makes it difficult to see the points in some cases; however, this can be read by seeing which fitted (blue) points fall outside the confidence interval. Residual plots show the standardized error for different predictor values, while partial regression plots show the predicted wildfire acres burned for different predictor values. The CCPR plots show just the coefficient multiplied by the predictor value for different predictor values, rather than the entire model fit.

Figure 6.2 shows the plots for the prescribed fires in 3 months and 50 km predictor. In the true versus fitted plot in the upper left corner, almost all true values fall within the confidence interval for the different values of prescribed burn

acreage. The residuals are mostly equally distributed, except for a vertical line on the far left. This pattern indicates a large number of wildfires had no prescribed burns in 3 months and 50 km, which results in a distribution of errors at that value. The residuals also appear more spread on the positive axis, which implies the model tends to overpredict the amount of wildfire acres burned when considering prescribed burns. The partial regression plot shows a slightly negative sloped line where increases in prescribed acres burned is associated with decreases in wildfire size. The CCPR plot again shows the horizontal line of wildfires with no prescribed fires nearby, yet the slight negative slope is still visible.

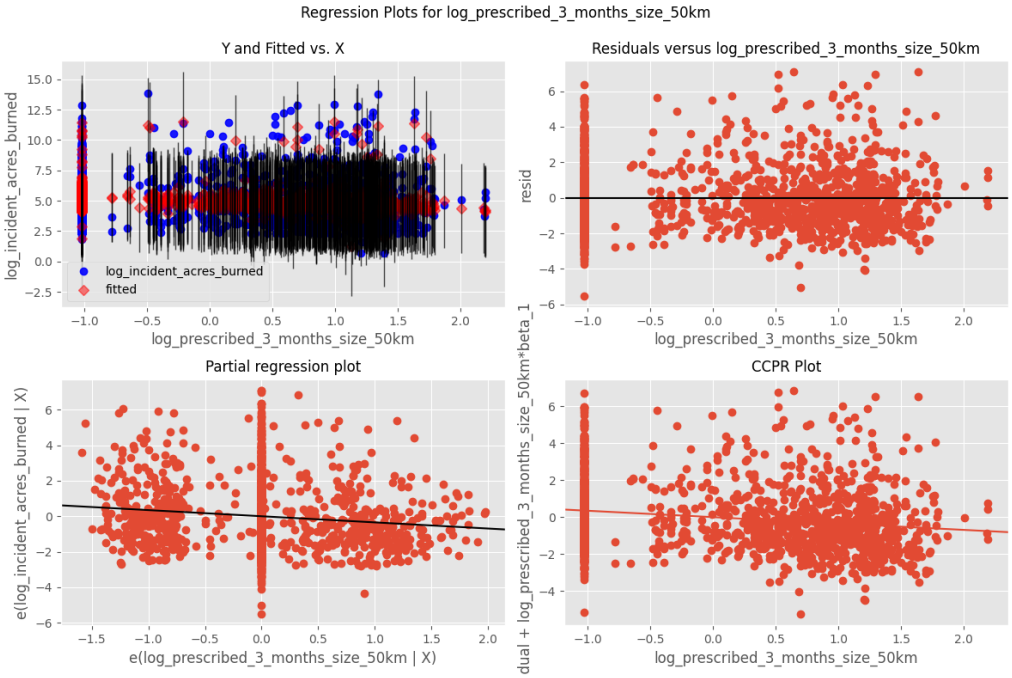


Figure 6.2: Regression results for prescribed fires in 3 months and 50 km.

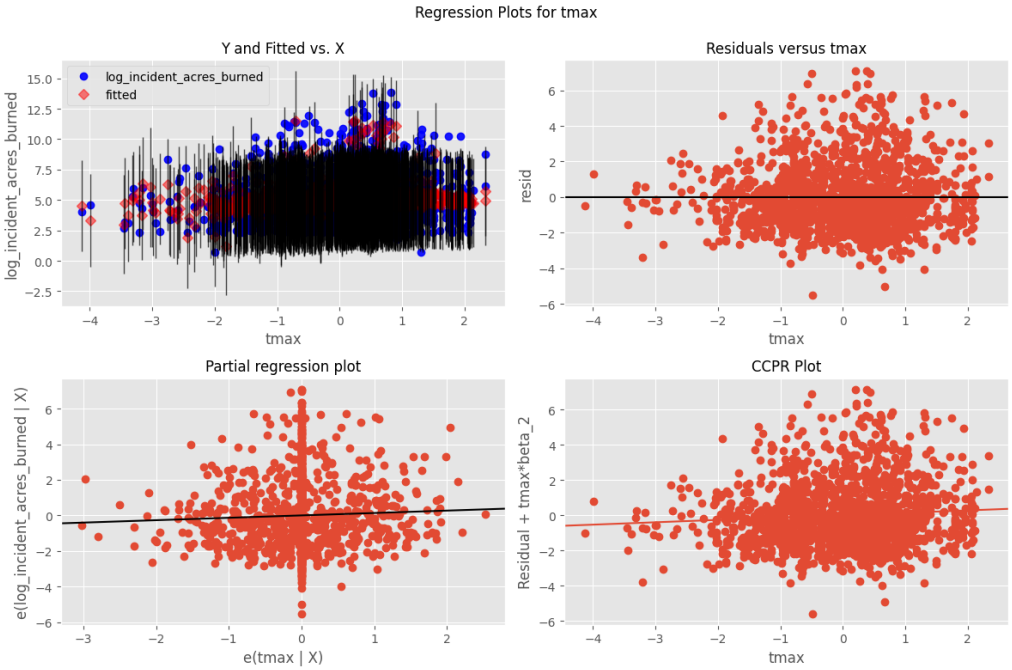


Figure 6.3: Regression results for maximum temperature.

Figure 6.3 shows similar results for maximum temperature. Again the true values are mostly contained in the fitted confidence intervals in the top left plot, though the true values appear to be at the top of the interval in some cases. The residuals in the top right appear to have greater variance for higher temperature values, with a greater distribution of high-magnitude positive residuals than negative residuals.In both the partial regression and CCPR plot, the slight positive relationship between maximum temperature and wildfire acres burned is visible.

Results for the precipitation predictor are available in figure 6.4. Here the fitted values also closely match the true values in the upper left corner plot, with only a few

data points having greater wildfire acres burned than predicted. The residual plot has more variation for low values of precipitation and little variation for high values of precipitation. This trend could indicate wildfires occuring after large amounts of precipitation have a more predictable outcome. Since most wildfires occur in the summer with little rain and generally vary in size, it makes sense that there would be more variation around the lower precipitation values. The partial regression plot shows a less steep negatively sloped line than those in the prescribed fire plots when the x-axis range is considered. This negative relationship is also visible in the CCPR plot.

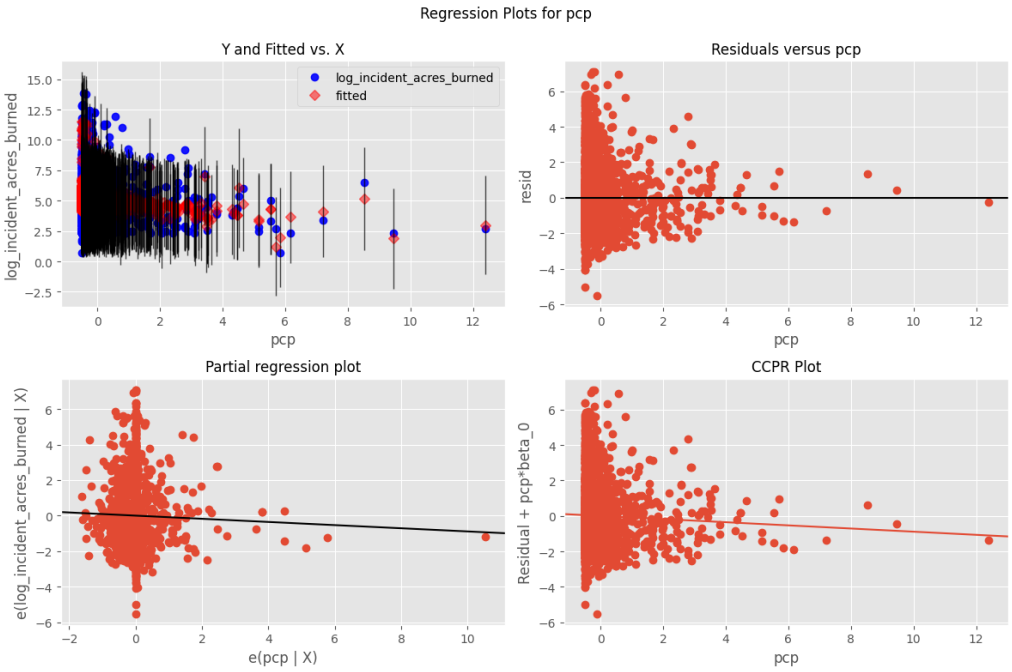


Figure 6.4: Regression results for precipitation.

Figure 6.5 shows the regression plots for the maximum wind speed variable. Similar to the other predictors, in the fitted versus true value plot the true values are almost entirely covered in the confidence intervals. The residuals are mostly evenly distributed, with a bit more clustering on the positive values, indicating an over-prediction of wildfire acres burned. There appears to be one outlier with a high wind speed, though the model fits it well with the residual close to zero. The partial regression plot shows a very slight negative trend, which is also visible in the CCPR plot.

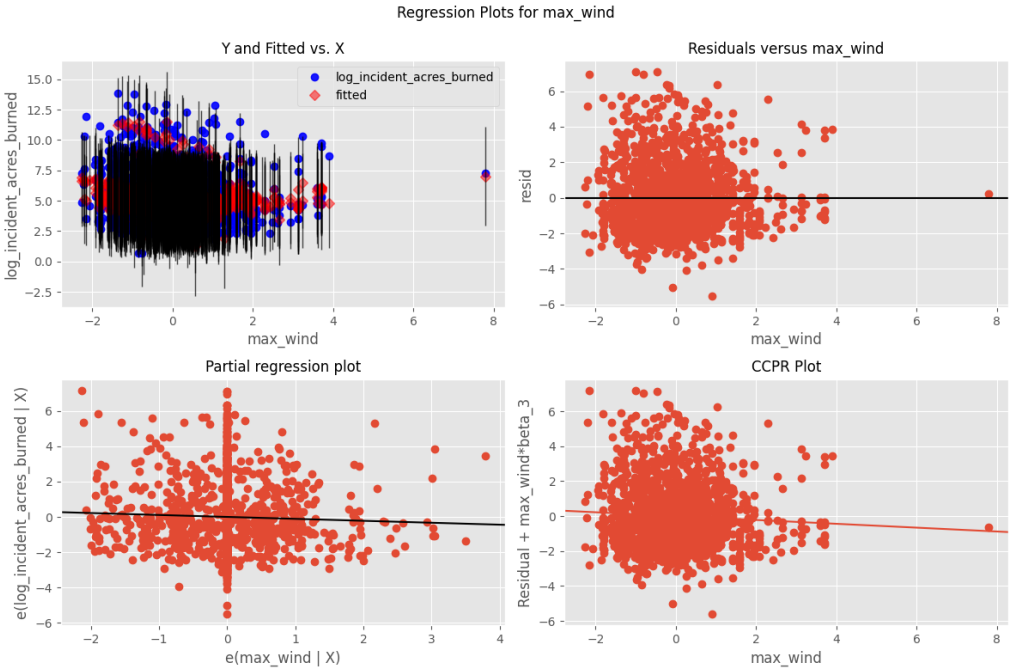


Figure 6.5: Regression results for maximum wind.

6.2.2 Linear Mixed-Effects Regression

Linear mixed-effect models are similar to regular linear regression models where the target is modeled by a linear combination of predictors, although mixed models allow for random intercepts from group variables and random slopes for predictors. The general formula for a linear mixed-effect model with p predictors can be seen in Equation (6.4), where fixed effects are denoted by β_n with $n \in [1, p]$, a random intercept is denoted by γ_0, and random slopes on variable m are denoted by γ_m with $m \in [1, p]$.

$$Y_i = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p + ... + \gamma_0 + \gamma_k x_k + ... + \gamma_l x_l + \epsilon_i \tag{6.4}$$

A linear mixed-effects model is also fitted on the four predictors (precipitation, maximum wind speed, maximum temperature, and prescribed fire acres within 3 months prior and 50 kms) with precipitation and prescribed burns as fixed effect variables, region as the grouping variable, and the random effect on the groups determined by a random intercept and a random slope for maximum temperature and precipitation. Results for the regression are in Table 6.6, where each fixed effect predictor is significant. Similar to the regular linear model, the fixed effect predictors all have a negative coefficient, indicating increases in the values are related to decreases in wildfire size. In contrast to the linear model, precipitation without interactions is a significant variable for predicting wildfire size.

Partial regression plots and residuals for the linear mixed-effect model are shown

Table 6.6: Linear mixed-effects regression results.

	Estimate	Std Error	z	P($>\|z\|$)	95% CI
***Intercept	4.796	0.311	15.397	0.000	[4.185, 5.407]
***pcp	-0.243	0.047	-5.140	0.000	[-0.336, -0.151]
***log_RX_3_50km	-0.248	0.051	-4.894	0.000	[-0.347, -0.149]
region Var	2.872	0.840			
region x tmax Cov	0.202				
tmax Var	0.025				
region x max_wind Cov	-0.593	0.207			
tmax x max_wind Cov	-0.067	0.011			
max_wind Var	0.185	0.061			
No. Observations:	1575	Method:	REML		
No. Groups:	8	Scale:	3.4463		
Min. group size:	34	Log-Likelihood:	-3237.9450		
Max. group size:	341	Converged:	Yes		
Mean group size:	196.9				

in Figures 6.6 and 6.7 respectively, with regions colored accordingly. Across the four predictors in Figure 6.6, fires in multiple regions are larger than the other regions, with a random intercept much higher than the other regions. The random slope for maximum temperature appears to result in two groupings between the regions, with the Central Coast and Shasta having flatter slopes than the other regions. This trend is also seen for the maximum wind speed, where the Central Coast and Shasta regions have a flatter, positive slope than the remaining regions. The residual plots in 6.7 show similar residual trends to the regular linear model, with all variables generally having more residuals with greater magnitudes on the positive axis than the negative axis.

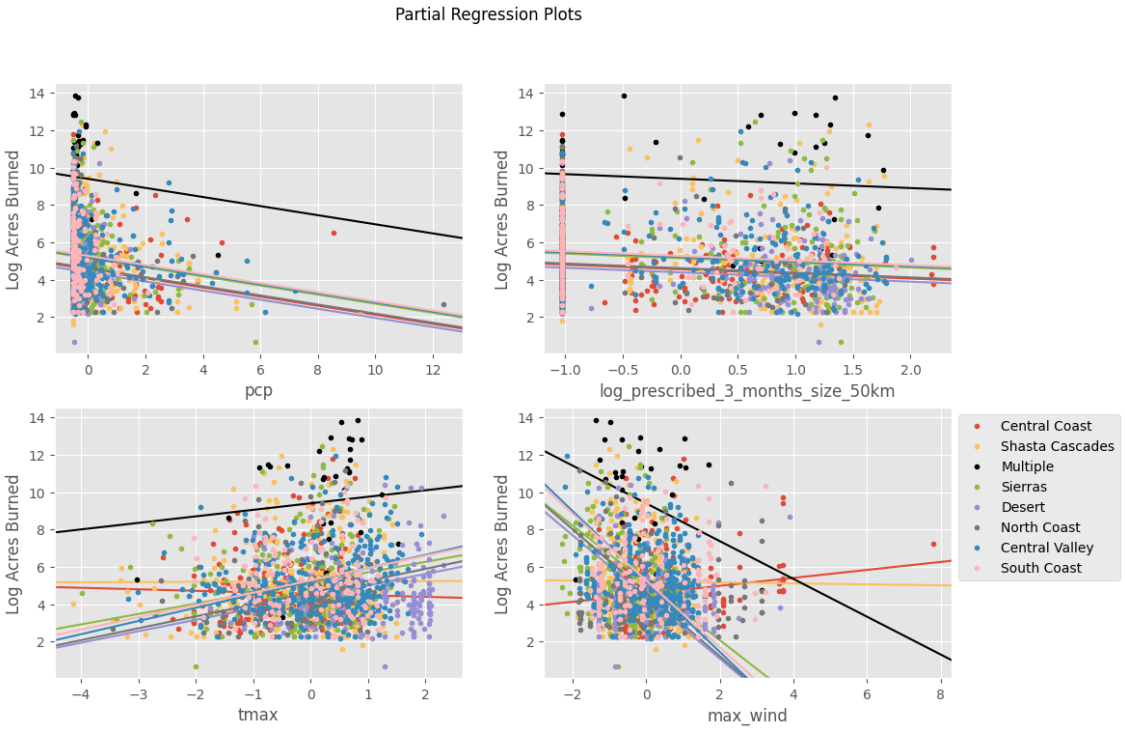


Figure 6.6: Partial regression plots for linear mixed model.

6.2.3 Non-parametric Regression

Non-parametric models use the data to fit a flexible model, where predictions are driven by the shape of the data rather than rigid parameters, like coefficients in linear regression. As a result, non-parametric models can fit complex data well and are considered for this regression problem. Here the K-nearest neighbors and support vector machine regressors are considered.

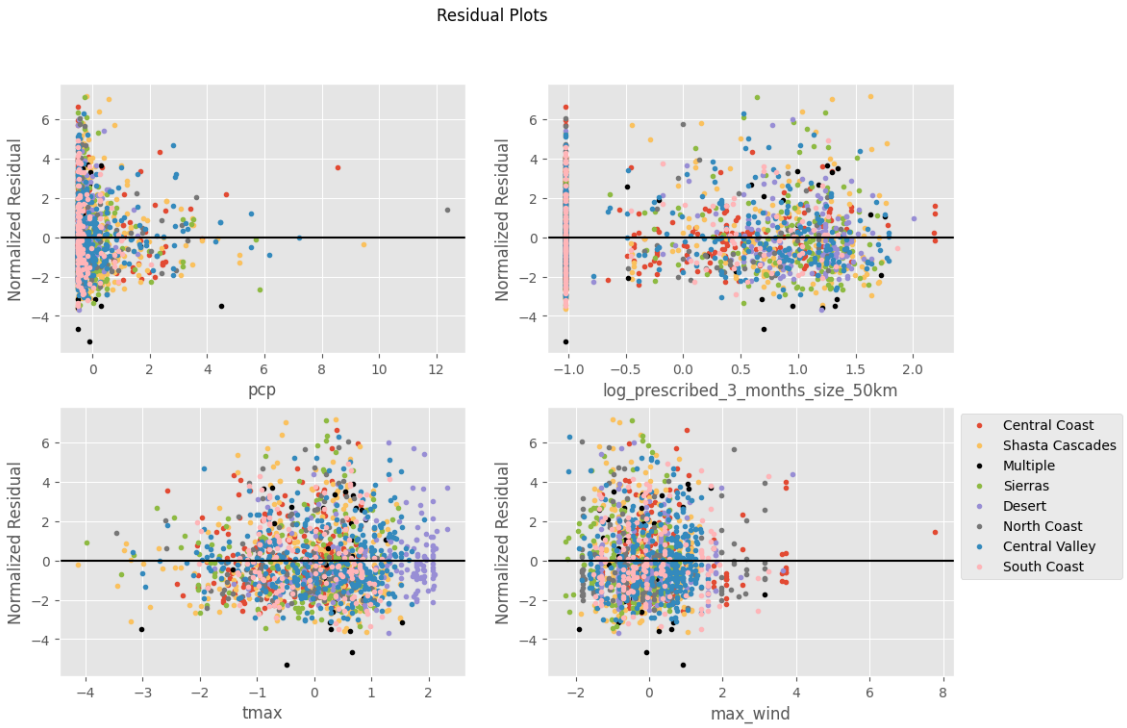


Figure 6.7: Residuals for linear mixed model.

K-Nearest Neighbors

The K-nearest neighbors (KNN) regression model predicts the target by identifying the k-closest events in the input space, then assigning the value to the new input based on the average of the values of the closest k inputs. The underlying assumption in KNN is that inputs resulting in similar scores on the target variable will be close together in the feature or input space. The best KNN model was found using cross-validation over the number of neighbors, k, with the values $k = 3, 5, 11, 15, 25$ considered, and the weighting of neighbors as uniform or based on distance. As a

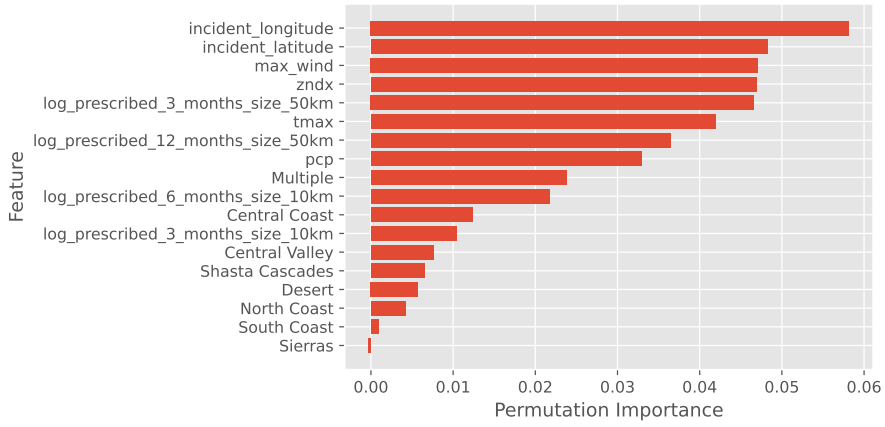


Figure 6.8: KNN feature importance.

result, the best model uses a uniform weight for neighbors and $k = 15$ neighbors. Predictor importance for the model is shown in Figure 6.8, where the top five predictors are longitude, latitude, maximum wind speed, drought index, and prescribed fires within 3 months and 50 km. The least important predictors appear to be the regions.

Support Vector Machines

When used for regression, support vector machines (SVMs) are similar to other regression models that fit a curve to the data. However, one key difference is the use of an allowed error boundary in the optimization of the curve fitting and the use of different kernels to fit the boundary. The best SVM model is found using cross-validation over the set of kernels (linear, polynomial, gaussian) and error boundaries

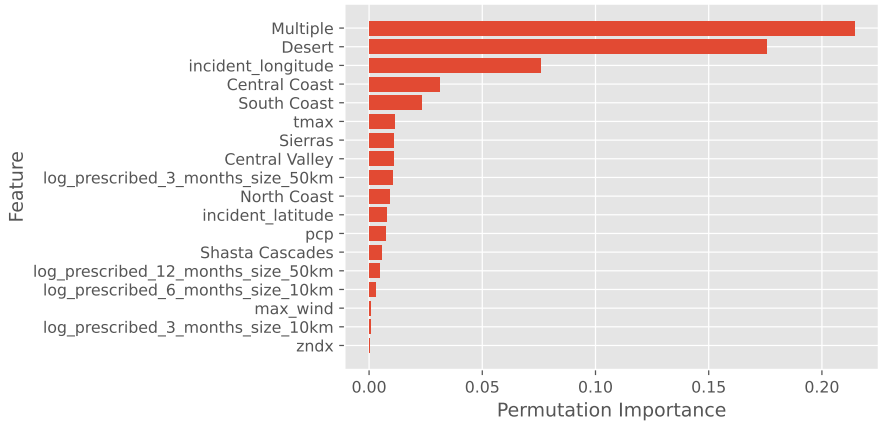


Figure 6.9: SVM feature importance.

($C = 0.001, 0.01, 0.1, 1, 10$), resulting in a model with a linear kernel and error boundary of $C = 1$. Predictor importance for the model is shown in Figure 6.9. In contrast to the KNN model, regions appear to be more important for the SVM, with fires occurring in Multiple regions as the most important predictor. Of the other predictors of interest, maximum temperature is most important, followed by prescribed fires within 3 months and 50 kms and precipitation.

6.2.4 Tree-based Regression

Two tree-based models are tested: random forest and AdaBoost. While Random Forest uses a bagging method to build an ensemble of trees from randomly selected subsets of features, AdaBoost uses boosting to recursively build new trees using the errors from older trees. Cross-validation is used to identify the best hyper-

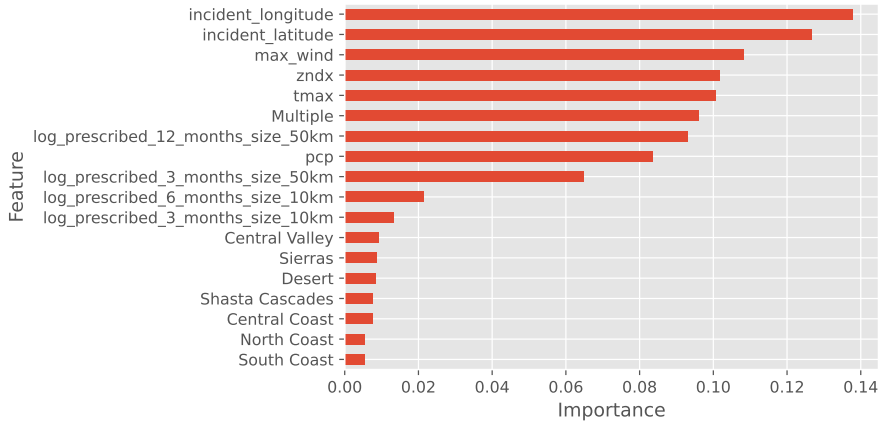


Figure 6.10: Random forest feature importance.

parameters for each model.

Random Forest

The random forest is fine-tuned for the number of trees (100, 250, 500, 750, 1000) and the number of predictors (p, $\sqrt{p}$, $log(p)$) are both tuned. As a result, the best Random Forest model has 1000 trees and *sqrtp* predictors. Feature importance is computed for the final model, with results in Figure 6.10. Similar to KNN, the random forest model finds longitude and latitude are important predictors, followed by maximum wind speed, maximum temperature, and drought index. The most important prescribed fire variable in this model is prescribed fires within 12 months and 50 km, followed by 3 monthss and 50 km.

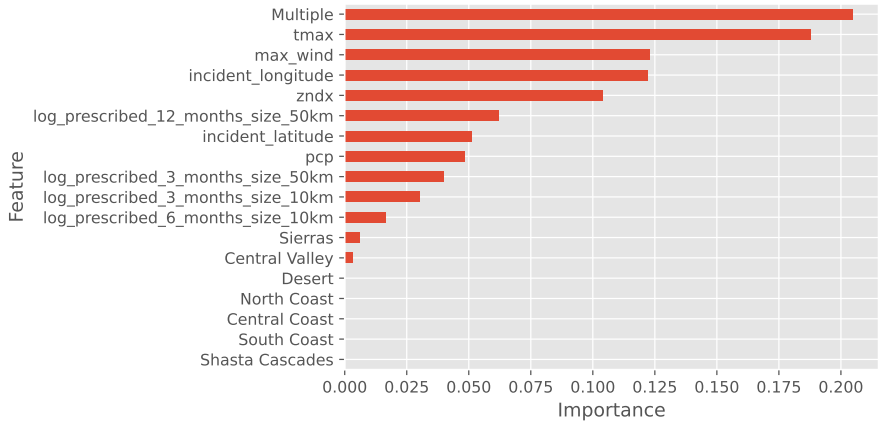


Figure 6.11: Adaboost feature importance.

AdaBoost

The AdaBoost model is only tuned on the number of trees (50, 100, 200, 300, 400, 500), with the best model having 50 trees. Feature importance is computed for the final model, with results in Figure 6.11. Similar to the SVM, the AdaBoost model finds fires occurring in multiple regions as the most important predictor. The climate predictors are more important than prescribed fire predictors in this model, with the drought index, maximum temperature, maximum wind speed, and precipitation all in the top eight most important predictors. The AdaBoost model agrees with the random forest model that prescribed fires within 12 months and 50 kms is the most important prescribed fire variable, though the other variables closely follow in importance ranking.

6.3 Model Comparison and Discussion

The performance of the six models is compared using MSE in Table 6.7 and Figure6.12 and by plotting the predicted versus true values in Figure 6.13. From the MSE comparisons in Table 6.7 and Figure6.12, the random forest clearly performs the best on both the training and test set. However, the large difference in MSE on the training (0.484) in comparison to the test set (4.216) indicates the random forest overfit to the training data. Despite the overfitting, this model still performs the best on the test set, though the other models are close behind. As expected, the null model performs the worst across training and testing, though the test MSE of 4.857 is actually an improvement on the mixed linear model's score of 5.066. Since the target variable of wildfire acres burned is transformed using the natural logarithm, the mean squared error of the random forest translates to a mean square error of 67.762 acres burned and a root mean squared error of 8.232 acres.

Table 6.7: Model MSE on train and test sets.

Model	Train MSE	Test MSE
null model	4.209	4.857
linear regression	3.396	4.467
linear mixed effects model	4.187	5.066
KNN	3.380	4.847
SVM	3.594	4.734
random forest	**0.484**	**4.216**
adaboost	4.006	4.749

The fit of each model is best seen with Figure 6.13. Most of the models seem

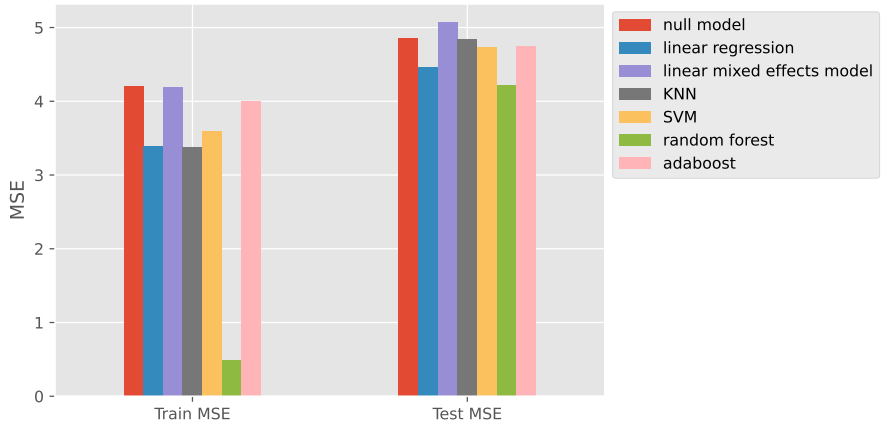


Figure 6.12: MSE on training and test sets.

to predict a fairly flat line, though the random forest has the most predictions that closely follow the true values. The linear model, AdaBoost, and SVM models also have some predictions that match the true values, especially on larger wildfires. One key consideration is the comparison of the null model to the fitted models. Overal the fitted models do not perform drastically better than the null, especially when comparing the MSEs in Table 6.7 and model output in Figure 6.13. This implies that while some information is gained from modeling wildfire size from the predictors, there is still a fair amount of variation unexplained by the models.

This regression analysis provides a method for quantifying the effect of prescribed burns on wildfire size while controlling for climate variables. While there is some variation among the models, the general trend is that both prescribed fires and climate variables are important for determining wildfire size, with the impact varying

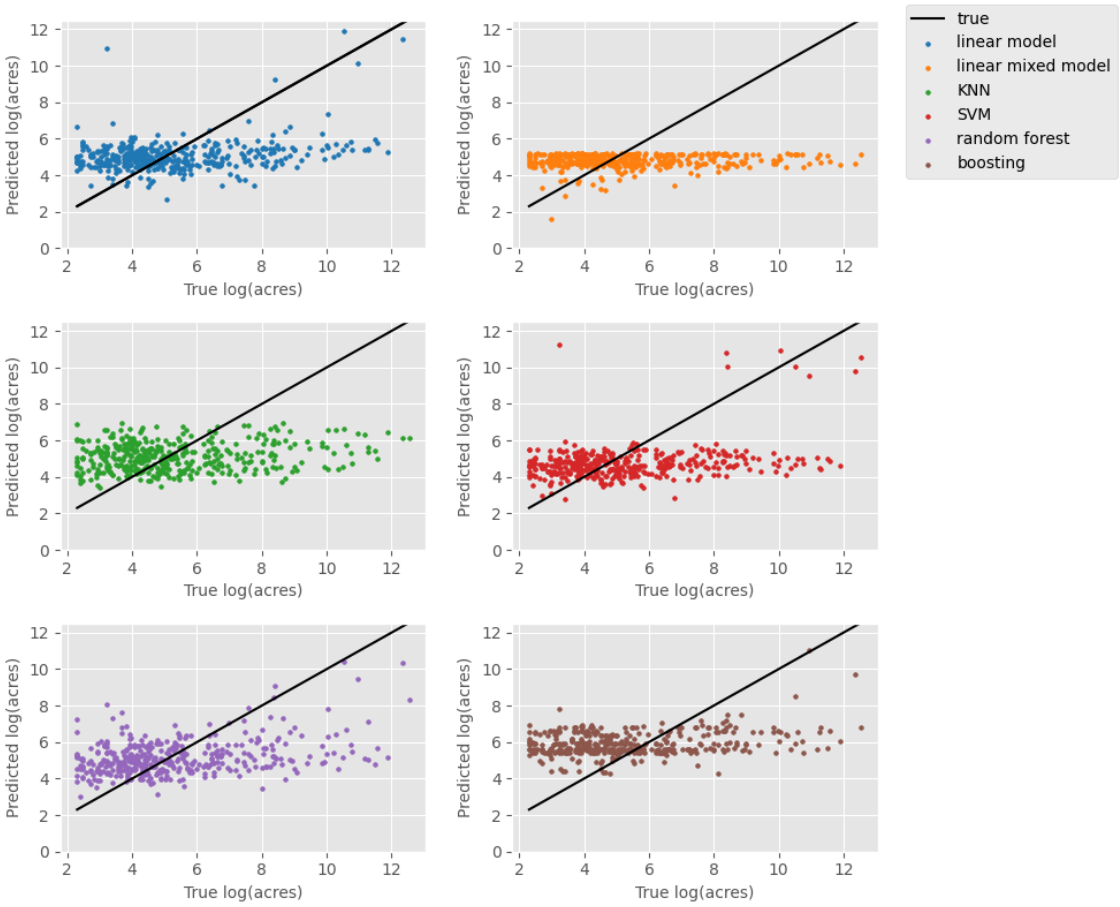


Figure 6.13: Model Performance

across spatial regions. Generally, precipitation, lower maximum temperature, wind speed, and prescribed fires are negatively related to wildfires, thus wildfires with greater precipitation, higher wind speeds, lower temperatures, and more prescribed fire acres burned nearby tend to be smaller when modeled using the methods in this section. Ultimately from this analysis prescribed fires do have a mitigating impact on wildfire size, even when controlling for the effect of climate variables.

Chapter 7

Discussion

The analyses provided throughout this study center around identifying which variables are related to wildfire size to ultimately inform mitigation strategies. While climate variables, like precipitation and temperature, are out of communities' control, prescribed fires are realistic control mechanism. Identifying at what spatial distance and temporal range prescribed fires are most effective could inform policy makers and forestry management personnel. Ultimately the results tend to point to prescribed fires within three months before a fire as most effective. This intuitively makes sense, as prescribed fires performed in the spring right before fire season would remove existing vegetation and new vegetation that grew since the previous fire season. In contrast, prescribed fires in fall would be performed prior to annual new growth and may be less effective. Another important consideration is the limited ability to implement prescribed burns. It is not feasible to simply burn all vegetation to prevent large wildfires, though that would have the greatest miti-

gating effect. Instead, prescribed burn placement should be optimized to maximize mitigation through proper spatial location.

There are a number of limitations to this research, with different benefits and drawbacks to each analysis. While the inferential statistics and times series analysis have limited assumptions, those models can only provide a pairwise relationship between wildfire size and one independent variable. In contrast, the regression analysis does allow for multiple predictor variables, though there are more stringent assumptions, such as Independence, that are required for a valid model. The linear regression model in particular is limited by the underlying data structures and is sensitive to colinearity among predictors. On the other hand, the linear model is valuable since it allows for inference on the predictor coefficients, resulting in a clear understanding of the relationships between predictors and wildfire size. Another key limitation is the treatment of spatial and temporal information in the prescribed burns variable. The analyses in this research rely on a binning method to engineer prescribed fire variables at different temporal and spatial ranges, rather than using an advanced integrated spatial-temporal model. Investigating advanced spatial-temporal models that allow for this type of modeling would be beneficial in future work.

Chapter 8

Conclusion

Throughout this research statistical tools have been used to analyze the impact of prescribed burns on wildfire size while considering climate variables. In particular, descriptive and inferential methods, time series analysis, and regression analysis were used to study what variables are related to wildfire size.

In the descriptive data analysis, general trends were identified, such as wildfires occurring more often in summer and prescribed fires occurring in fall, winter in spring. Inferential statistics found wildfires with a prescribed fire near by tend to be smaller on average than wildfires without one, providing evidence for prescribed fire's efficacy. Tests also showed wildfires with average temperature greater than sixty degrees Fahrenheit, monthly precipitation less than one inch, and negative drought indices are larger on average. Additional inferential tests found the distribution of wildfires and prescribed fires varies across regions.

The time series analysis found that climate variables, including temperature

and precipitation, have a stronger relationship with wildfire acres burned when considering monthly sums across the state. From the cross-correlation time series analysis, climate variables in the month of the fire have the strongest correlation to wildfire acres burned.

The regression analysis predicted wildfire size from region, climate, and prescribed fire variables, in turn providing a measure of the effect of prescribed burns while controlling for climate variables. Linear models, KNN, and SVM all found that prescribed burns have the strongest effect when performed three months prior and within 50 km from a wildfire, while the tree based models found prescribed burns most effective at twelve months prior and 50 km from a wildfire. The models agreed wind, precipitation, and temperature were important climate variables that impact wildfire size. Additionally, region and fire location were important predictors across models as well.

Overall, the analyses find that prescribed fires are related to decreased wildfire size, even when controlling for climate conditions. These findings help support the implementation of prescribed burns in California, despite varying climate conditions. With wildfire destruction increasing in recent years and climate projections predicting severe future fires, identifying potential mitigation methods is critical to limit future devestation. However, prescribed burns are only a small piece of the puzzle and other components, such as community resilience, first responder capabilities, and rapid detection are also possible mechanisms for combating future fires.

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